



THE UNIVERSITY OF
MELBOURNE

Probability Reliability and Quality Regression

Kevin Otto

Schedule and Learning Outcomes

Week	Recorded Lecture Topic
1	Introduction.
2	Inspection. Yield.
3	ANOVA. Measurement Systems Analysis.
4	Statistics. Confidence Intervals.
5	Hypothesis Testing. Sample Sizes.
6	Quality Management Processes.
7	Regression.
8	Design of Experiments.
9	System Reliability. Availability.
	<i>BREAK – Non Teaching Week</i>
10	Parametric Distribution Analysis
11	Statistical Process Control. Supplier Quality.
12	Review

After completion of the course the student should be able to:

- Explain different statistical methods
- **Parametric and non-parametric hypothesis tests**
- Design and plan for collecting experimental data
- Concepts of reliability engineering and the relationship between TTF, R, H
- **How to improve its processes, products and services**
- Apply TQM, 6S, and benchmarking



Overview

OLS regression

Regression Table

Residual Analysis

Polynomial Regression

Multiple Regression

Over Fitting

Maximum Likelihood Regression



Regression

Method to determine the statistical relationship between a response (or output) and one or more predictor (or input) variables.

$$Y = f(X_1, \dots, X_n)$$

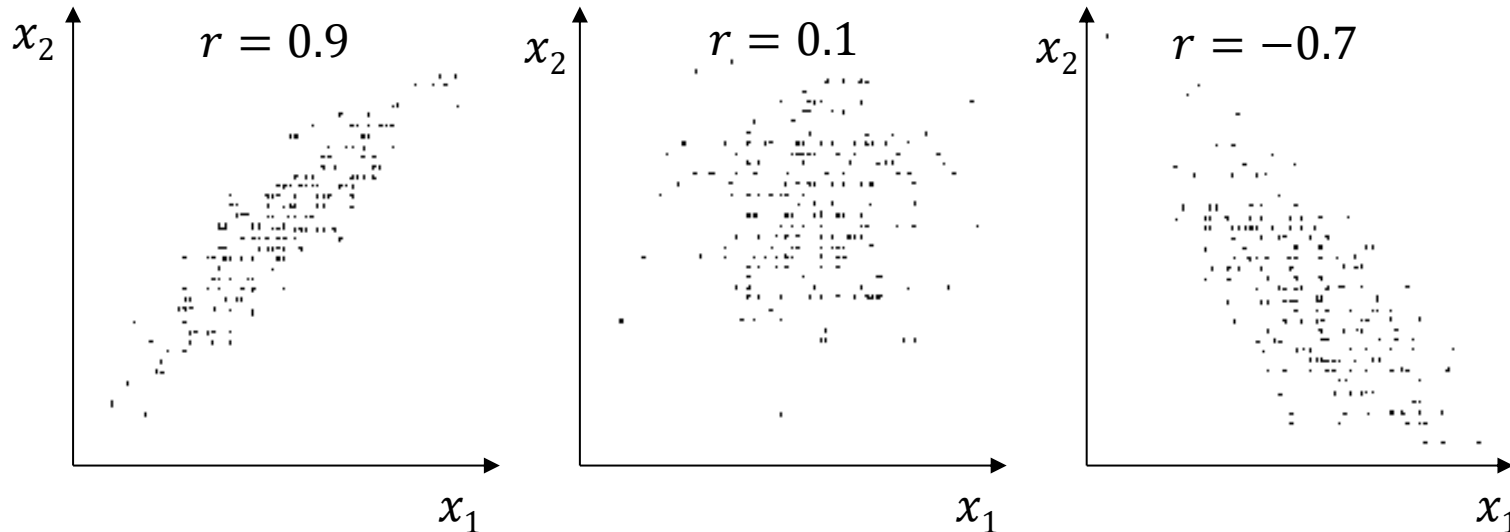
The diagram shows the equation $Y = f(X_1, \dots, X_n)$ with two red arrows pointing upwards. One arrow points from the word 'response' to the variable Y . The other arrow points from the word 'predictors' to the variables $X_1, \dots, X_n$.

response predictors

Regression analysis develops an estimating equation, i.e. a formula f that relates the predictor(s) to the response.

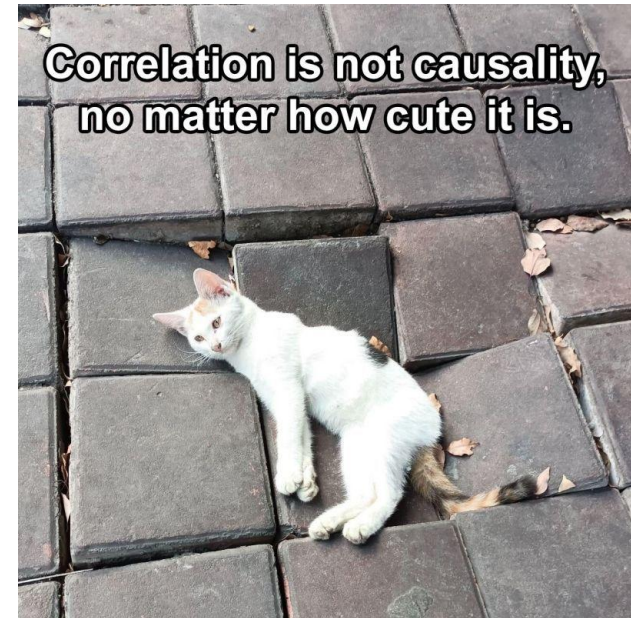
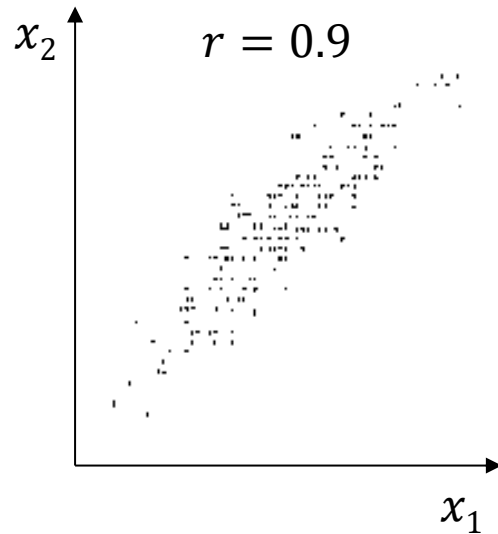
Coefficient of Correlation

The ***Coefficient of Correlation*** (r), measures the strength of a linear relationship between the response and a single predictor.



Coefficient of Correlation

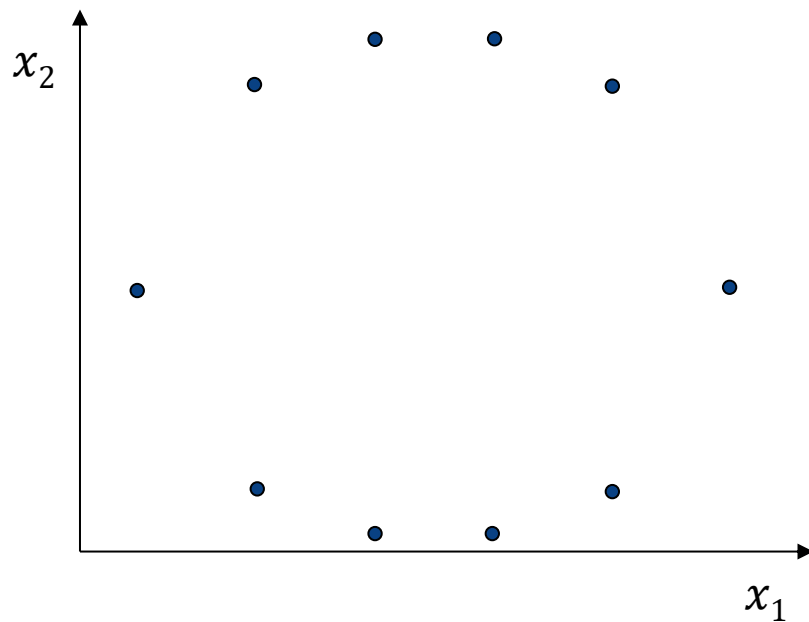
High correlation r doesn't mean x_2 is caused by x_1 .



$$r = 0$$

$r = 0$ implies no linear relationship.

$r = 0$ does not imply no relationship!

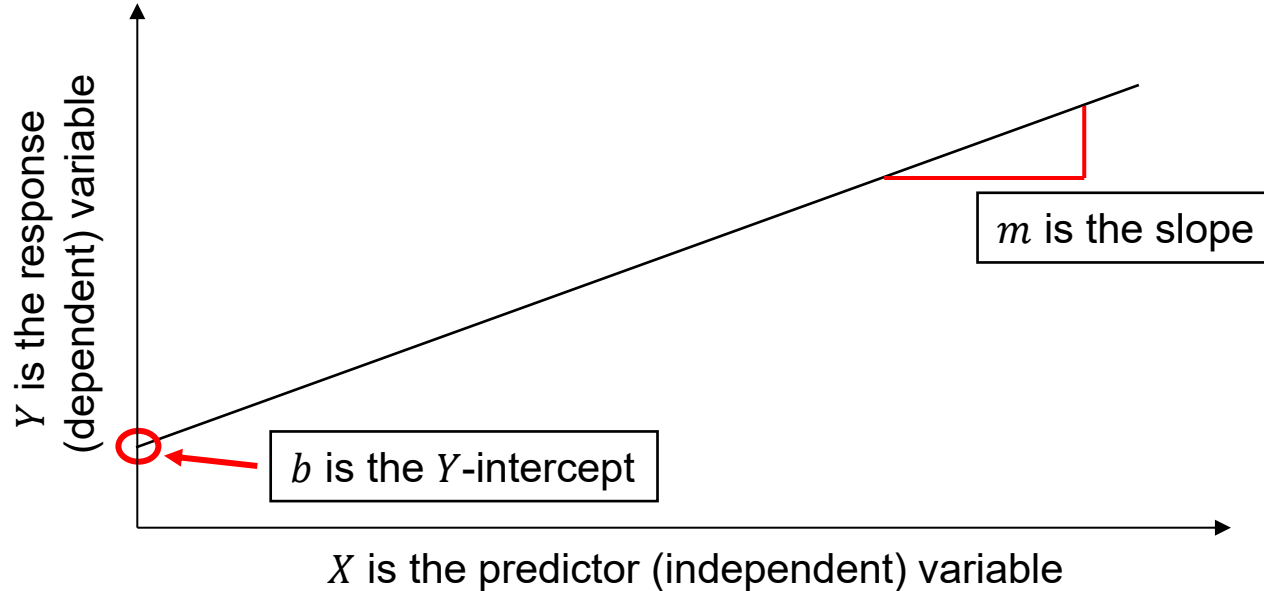


Ordinary Least Squares Linear Regression

The simple linear regression model is

$$Y = mX + b$$

where:





Assumptions of Regression

In order for regression analysis to work, certain assumptions must be made about the data...

A true model actually exists, and it takes the form

$$Y = mX + b$$

- The “ Y ” values are independent of each other
- The variability seen in Y (σ_Y) is the same at any X , it is constant
- The variability should not be significantly non-normal
- There is no error in X ($\sigma_X = 0$)

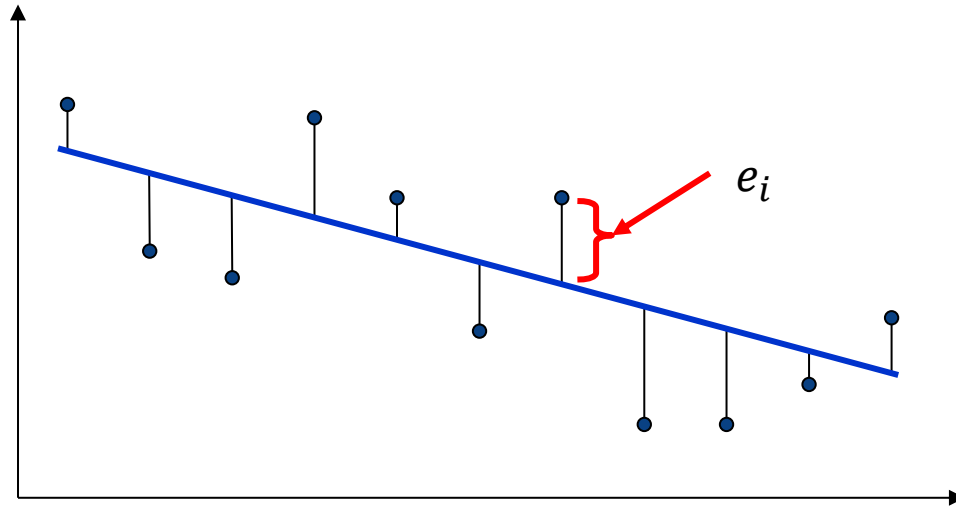
Method of Least Squares

The residual (or error) is denoted by

$$e_i = Y_{\text{observed } i} - Y_{\text{predicted } i}$$

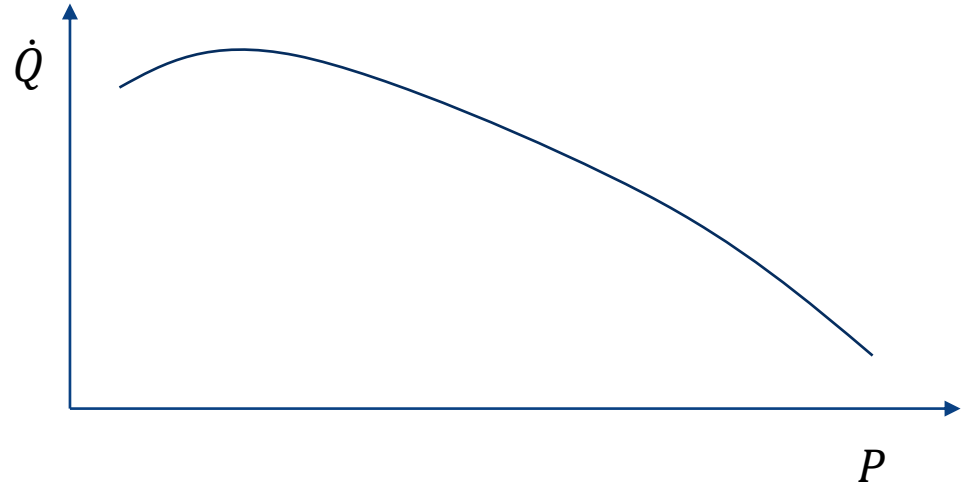
The best fitting straight line is the line for which the sum of the squared residuals is minimized. That is, the values for the (m, b) statistics such that

$$(m, b)_* = \operatorname{argmin}_{m, b} \sum e_i^2$$



Simple Linear Regression Example

Fan curves are created to parameterize airflow as a function of pressure drop.





Simple Linear Regression Example

Fan curves are created to parameterize SCFM as a function of pressure drop.

- These curves are determined from sweeps done in the lab

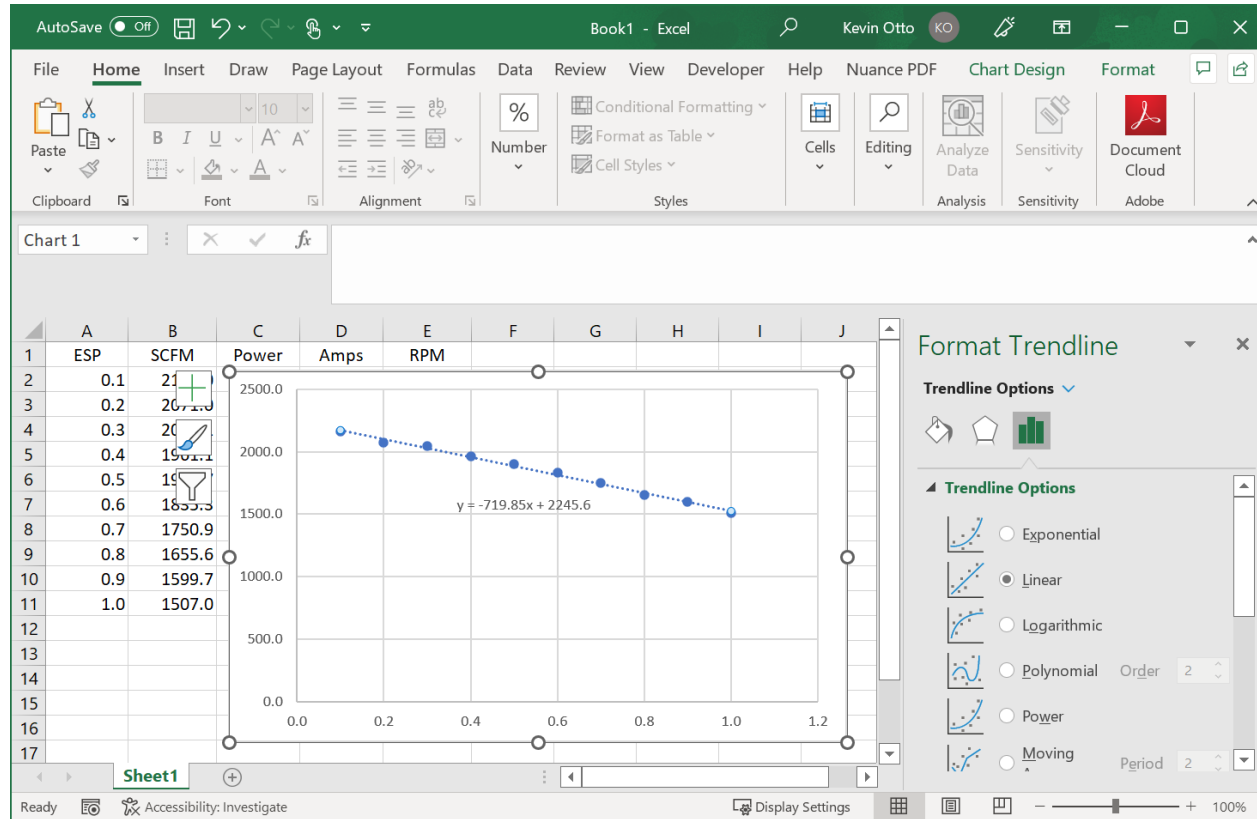
Determine the linear regression model of SCFM in terms of static pressure drop.

$$SCFM = \beta_0 + \beta_1 ESP$$

ESP	SCFM	Power	Amps	RPM
0.1	2165.0	596.7	8.452	854.9
0.2	2071.0	608.9	8.619	880.5
0.3	2047.1	621.2	8.783	910.4
0.4	1961.1	634.6	8.961	935.3
0.5	1903.7	648.7	9.140	963.0
0.6	1835.3	661.7	9.308	991.7
0.7	1750.9	670.8	9.425	1018.5
0.8	1655.6	682.3	9.574	1047.5
0.9	1599.7	689.9	9.665	1076.5
1.0	1507.0	700.1	9.798	1107.0

Fitted Line Regression – Excel

In Excel create a Scatter plot. Then add a trendline using options.



Fitted Line Regression – Minitab

A regression model for SCFM vs ESP:

Stat > Regression > Fitted Line Plot...

Fitted Line Plot

Response (Y): SCFM

Predictor (X): ESP

Type of Regression Model

☒ Linear ☐ Quadratic ☐ Cubic

Select

Graphs...

Options...

Storage...

Help

OK

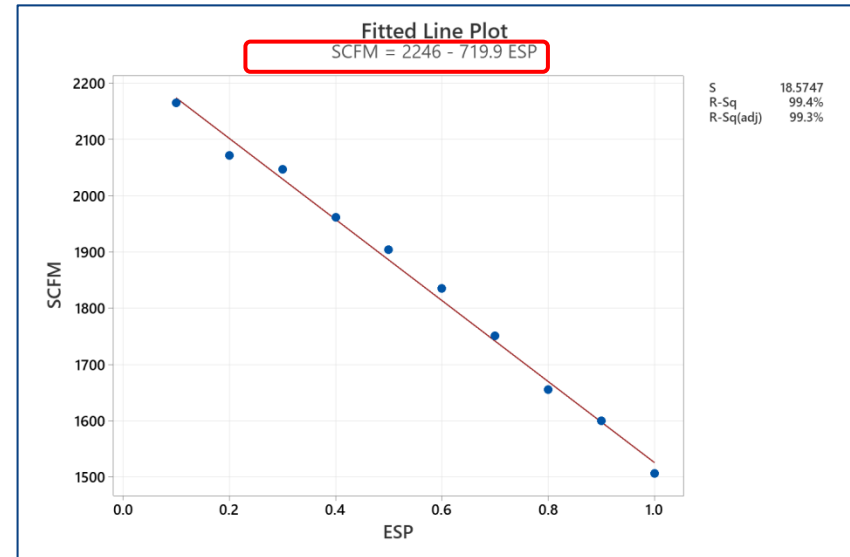
Cancel

Model Summary

S	R-sq	R-sq(adj)
18.5747	99.36%	99.28%

Analysis of Variance

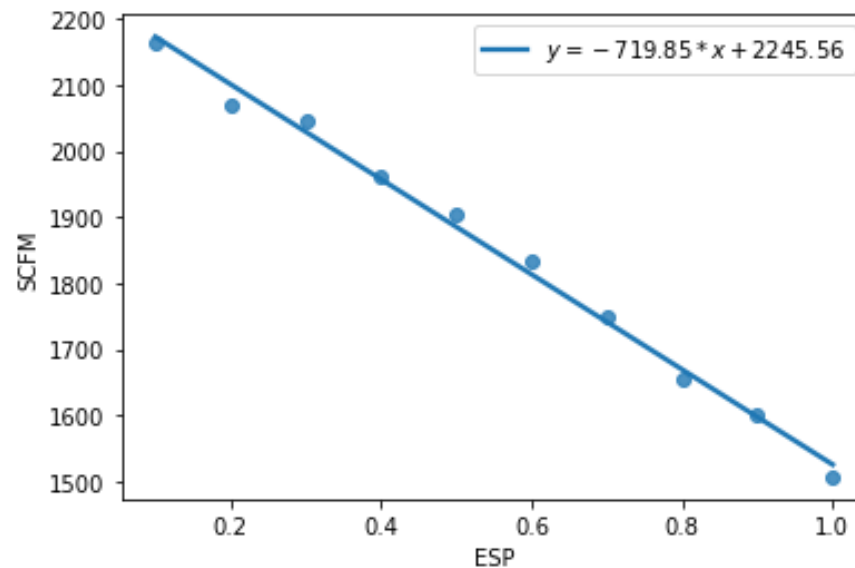
Source	DF	SS	MS	F	P
Regression	1	427507	427507	1239.09	0.000
Error	8	2760	345		
Total	9	430267			



Fitted Line Regression – Python

```
lfit = df.copy()
lreg = ols('SCFM ~ ESP', data=lfit).fit()
intercept = lreg.params['Intercept']
slope = lreg.params['ESP']

with Figure(4, 3) as (fig, ax):
    sns.regplot(x='ESP', y='SCFM', data=df, ci=0, ax=ax)
    ax.legend([f'$y = {slope:.2f} * x + {intercept:.2f}$'])
```





Least Squares Regression: Theory

Find equation f which best fits the data.

$$y = f(\bar{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

At each experimental point

$$y_{\text{measured}} = y_{\text{predicted}} + \varepsilon = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

A measure of the model error is the squared sum

$$\begin{aligned}\varepsilon^2 &= \sum_h \varepsilon_h^2 \\ &= \sum_h (y_{\text{measured}} - y_{\text{predicted}})^2 \\ &= \sum_h (y_h - \beta_0 - \beta_1 x_{1h} - \beta_2 x_{2h} - \beta_3 x_{3h})^2\end{aligned}$$



Least Squares Regression

One definition of the best β s are the ones that minimize ε^2

This is where the derivative of ε^2 is zero with respect to each β

$$\frac{\partial \varepsilon^2}{\partial \beta_i} = 0$$

$$\frac{\partial \varepsilon^2}{\partial \beta_0} = 0 = -2 \sum_h (y_h - \beta_0 - \beta_1 x_{1h} - \beta_2 x_{2h} - \beta_3 x_{3h})$$

$$\frac{\partial \varepsilon^2}{\partial \beta_i} = 0 = -2 \sum_h (y_h - \beta_0 - \beta_1 x_{1h} - \beta_2 x_{2h} - \beta_3 x_{3h}) x_{ih}$$

Least Squares Regression

Rearranging in matrix form

$$\begin{bmatrix} N & \sum x_{1h} & \sum x_{2h} & \sum x_{3h} \\ \sum x_{1h} & \sum x_{1h}^2 & \sum x_{1h}x_{2h} & \sum x_{1h}x_{3h} \\ \sum x_{2h} & \sum x_{1h}x_{2h} & \sum x_{2h}^2 & \sum x_{2h}x_{3h} \\ \sum x_{3h} & \sum x_{1h}x_{3h} & \sum x_{2h}x_{3h} & \sum x_{3h}^2 \end{bmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} \sum y_h \\ \sum y_h x_{1h} \\ \sum y_h x_{2h} \\ \sum y_h x_{3h} \end{pmatrix}$$

That is,

$$[A]\vec{\beta} = \vec{B}$$

And so a closed form solution exists as

$$\vec{\beta} = [A]^{-1}\vec{B}$$

Linear Algebra

Define the design matrix as a matrix X .

Run	x_1	x_2	x_3
1	1	1	1
2	1	1	-1
3	1	-1	1
4	1	-1	-1
5	-1	1	1
6	-1	1	-1
7	-1	-1	1
8	-1	-1	-1

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & -1 \end{bmatrix}$$

Linear Algebra

Then one can write

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & -1 \end{bmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \\ y_8 \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \\ \varepsilon_7 \\ \varepsilon_8 \end{pmatrix}$$

And so more simply,

$$X\vec{\beta} = \vec{y} + \vec{\varepsilon}$$



Linear Algebra Derivation

With

$$X\vec{\beta} = \vec{y} + \vec{\varepsilon},$$

Pre-multiply by X^T :

$$X^T X \vec{\beta} = X^T \vec{y} + X^T \vec{\varepsilon},$$

This is invertible

$$\vec{\beta} = [X^T X]^{-1} X^T \vec{y}$$

Further the $[X^T X]^{-1} X^T \vec{\varepsilon}$ is an error term on the coefficients and can be used to construct confidence intervals and p-values on the best-fit coefficients $\vec{\beta}$.



Regression

Practically, we compute this analysis using statistical analysis software

- Excel regression
- Minitab
- Python `scipy.stats`, `pandas`, `sklearn`, `seaborn`, ...
- JMP
- R
- Statistica
- Matlab statistics toolbox
- etc.

Example: Fan Curves

Fan curves are created to parameterize SCFM as a function of pressure drop.

- These curves are determined from sweeps done in the lab

Determine the linear regression model of SCFM in terms of static pressure drop.

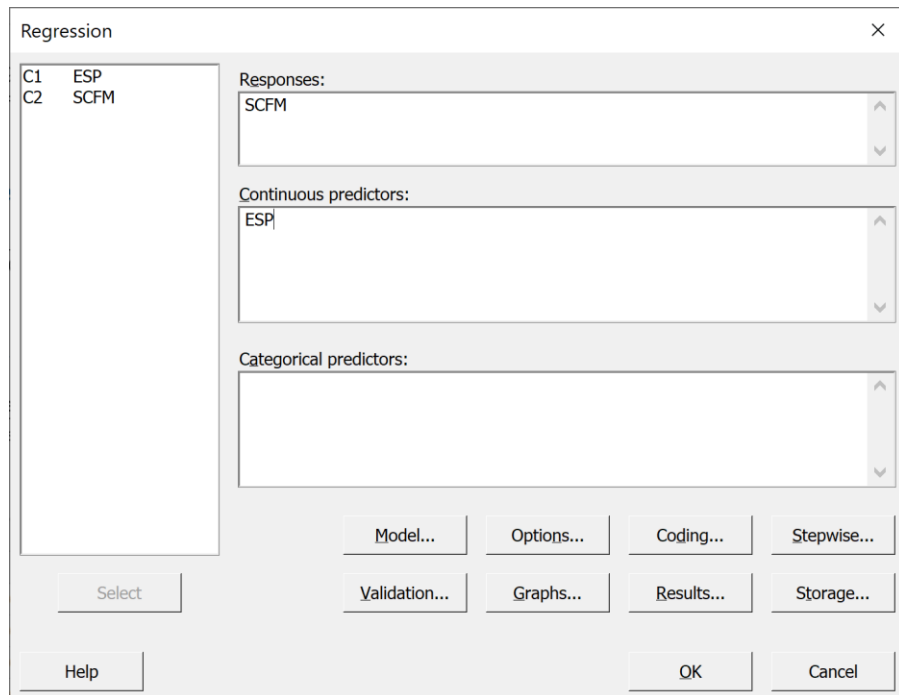
$$SCFM = \beta_0 + \beta_1 ESP$$

ESP	SCFM
0.1	2165.0
0.2	2071.0
0.3	2047.1
0.4	1961.1
0.5	1903.7
0.6	1835.3
0.7	1750.9
0.8	1655.6
0.9	1599.7
1.0	1507.0

Simple Linear Regression – Minitab

A regression model for SCFM vs ESP:

Stat > Regression > Regression > Fit Regression Model...



The Minitab Regression dialog box is shown. The 'Responses' field contains 'SCFM'. The 'Continuous predictors' field contains 'ESP'. The 'Categorical predictors' field is empty. The 'Model...' button is highlighted.

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
18.5747	99.36%	99.28%	98.95%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	427507	427507	1239.09	0.000
ESP	1	427507	427507	1239.09	0.000
Error	8	2760	345		
Total	9	430267			

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	2245.6	12.7	176.97	0.000	
ESP	-719.9	20.5	-35.20	0.000	1.00

$$SCFM = 2245.6 - 719ESP$$

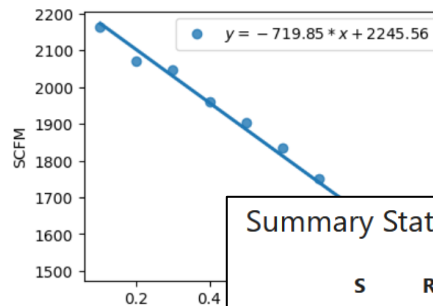
Simple Linear Regression – Python

```
data = {'ESP':[0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0],
        'SCFM':[2165.0, 2071.0, 2047.1, 1961.1, 1903.7,
                1835.3, 1750.9, 1655.6, 1599.7, 1507.0]}
df = pd.DataFrame(data)

lfit = df.copy()
lreg = ols('SCFM ~ ESP', data=lfit).fit()
intercept = lreg.params['Intercept']
slope = lreg.params['ESP']

with Figure(4, 3) as (fig, ax):
    sns.regplot(x='ESP', y='SCFM', data=df, ci=0, ax=ax)
    ax.legend(['f'$y = {slope:.2f} * x + {intercept:.2f}$'])

vstack(
    '### Summary Statistics',
    mqr.anova.adequacy(lreg),
    '### ANOVA of the Regression',
    mqr.anova.summary(lreg),
    '### Regression Table',
    mqr.anova.coefls(lreg))
```



Summary Statistics

S	R-sq	R-sq (adj)	N	JB	p (JB)
18.574669	0.993585	0.992783	10	0.688977	0.708583

ANOVA of the Regression

	df	sum_sq	mean_sq	F	PR(>F)
ESP	1.0	427507.217455	427507.217455	1239.085564	4.642484e-10
Residual	8.0	2760.146545	345.018318	NaN	NaN
Total	9.0	430267.364000	427852.235773	NaN	NaN

Regression Table

	Coeff	[2.5%	97.5%]	PR(>F)	VIF
Intercept	2245.560000	2216.299311	2274.820689	1.163100e-15	4.666667
ESP	-719.854545	-767.012416	-672.696675	4.642484e-10	1.000000

Simple Linear Regression – Excel

A regression model for SCFM vs ESP:

Data > Data Analysis > Regression

Regression

Input

Input Y Range:

Input X Range:

☒ Labels ☐ Constant is Zero

☐ Confidence Level: %

Output options

☒ Output Range:

☐ New Worksheet Ply:

☐ New Workbook

Residuals

☐ Residuals ☐ Residual Plots

☐ Standardized Residuals ☐ Line Fit Plots

Normal Probability

☐ Normal Probability Plots

OK Cancel Help

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.997				
R Square	0.994				
Adjusted R Square	0.993				
Standard Error	18.6				
Observations	10				
ANOVA					
	df	SS	MS	F	Sig F
Regression	1	427507.2	427507.2	1239.1	0.000
Residual	8	2760.1	345.0		
Total	9	430267.4			
	Coefficients	Std Error	t Stat	P-value	
Intercept	2245.6	12.69	177.0	0.000	
ESP	-719.9	20.45	-35.2	0.000	



Overview

OLS regression

Regression Table

Residual Analysis

Polynomial Regression

Multiple Regression

Over Fitting

Maximum Likelihood Regression

Is the Equation Any Good?

The regression statistics.
What do they mean?

SUMMARY OUTPUT					
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	Coefficients	Std Error	t Stat	P-value	
Intercept	2245.6	12.69	177.0	0.000	
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Coefficient of Determination

The **Coefficient of Determination** (r^2) measures the proportion of variability in Y which can be explained by Y 's relationship with X in the derived model $Y = mX + b$

$$r^2 = \frac{SS_{\text{regression}}}{SS_{\text{total}}}$$

A higher value for the coefficient of determination implies less error between the regression model and the observed values.

- For single variable problems $r^2 = (\text{Coefficient of Correlation})^2$

Adjusted r^2

The **adjusted r^2** is a modified r^2 that also considers the number of predictor variables included in the regression equation:

$$r_{\text{adjusted}}^2 = 1 - \left(\frac{n - 1}{n - p - 1} \right) (1 - r^2)$$

where n = # data points, p = # predictor terms (x's)

While addition of terms will always cause r^2 to increase, the r_{adjusted}^2 will decrease if the added terms are insignificant.

RMS Error

The ***standard error*** s of the regression is

$$s = \sqrt{\frac{\text{Sum of Squared Errors}}{\text{Degrees of Freedom}}}$$

It is also called the RMSE or Root Mean Square Error of the model fit. It can be computed by taking the square root of the Mean Square of Errors in the ANOVA table of the regression.

Is the Equation Any Good?

In the regression statistics output:

r^2
 r^2_{adjusted}
 Standard error s

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.997				
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Coefficients					
	Coefficients	Std Error	t Stat	P-value	
Intercept	2245.6	12.69	177.0	0.000	
ESP	-719.9	20.45	-35.2	0.000	

ANOVA of the Regression

The ANOVA table compares the prediction sum of squares with the residual error sum of squares.

- The null hypothesis is $SS_{\text{regression}} = 0$

This p-value should be small.

If not, $SS_{\text{regression}} \ll SS_{\text{residual}}$, meaning the regression is no good.

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.997				
R Square	0.994				
Adjusted R Square	0.993				
Standard Error	18.6				
Observations	10				
ANOVA					
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Total	9	430267.4			
	Coefficients	Std Error	t Stat	P-value	
Intercept	2245.6	12.69	177.0	0.000	
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$$p = Fdist(F, df_{reg}, df_{res})$$

$$F = MS_{reg}/MS_{res}$$

$$MS_i = SS_i/SS_{res}$$

The Regression Table

The regression table lists the model coefficients.

The p-values are the probability the coefficients arose randomly and should in fact be zero.

The p-values of each term should be small. If not, the term should be dropped from the equation.

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.997				
R Square	0.994				
Adjusted R Square	0.993				
Standard Error	18.6				
Observations	10				
ANOVA					
	df	SS	MS	F	Sig F
Regression	1	427507.2	427507.2	1239.1	0.000
Residual	8	2760.1	345.0		
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	Coefficients	Std Error	t Stat	P-value	
Intercept	2245.6	12.69	177.0	0.000	
ESP	-719.9	20.45	-35.2	0.000	

$$p = tdist(t, df_{res})$$

$$t_i = coef_i / se_i$$

Regression – p value interpretation

Recall that $Y = mX + b$ (expressed as sample statistics)

or

$$Y = \beta_0 + \beta_1 X \quad (\text{expressed as population parameters})$$

Hypothesis (Constant): $H_0: \beta_0 = 0, H_A: \beta_0 \neq 0$

- Implications: If β_0 is 0 (Accept H_0), then the regression line runs through the origin ($x = 0, y = 0$), and does not contribute significantly to the prediction of Y .

Hypothesis (X): $H_0: \beta_1 = 0, H_A: \beta_1 \neq 0$

- Implications: If β_1 is 0 (Accept H_0), then X can be ANY value and Y will not be affected. Consequently X does not contribute significantly to the prediction of Y .



Common Regression Mistakes

Excessive focus on the Coefficient of Correlation

Attributing cause and effect

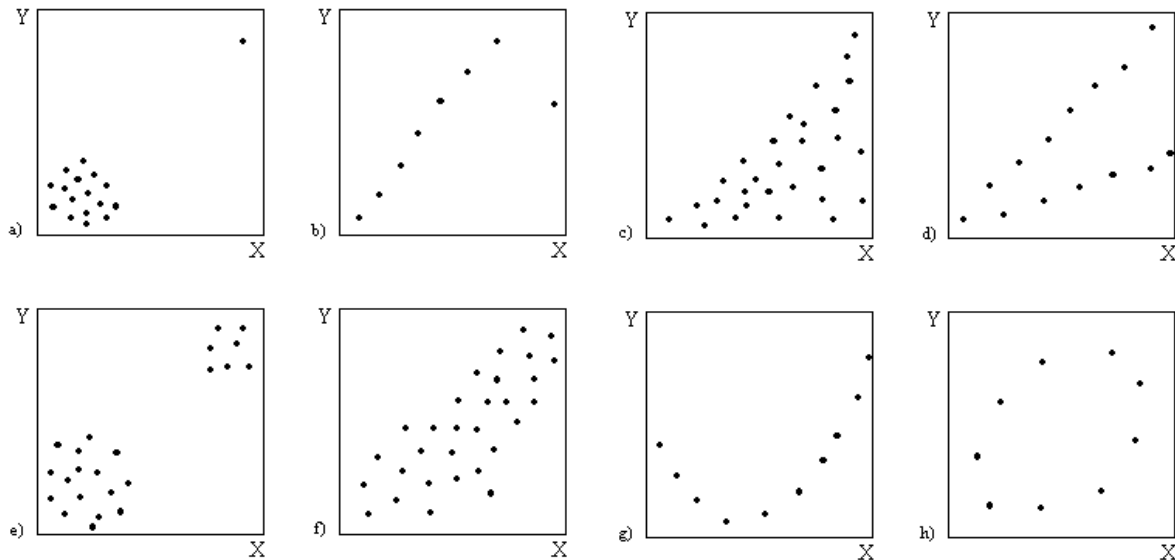
Extrapolating the relationship beyond the data range

Masking real correlations or creating false correlations

Data collected over too narrow a range

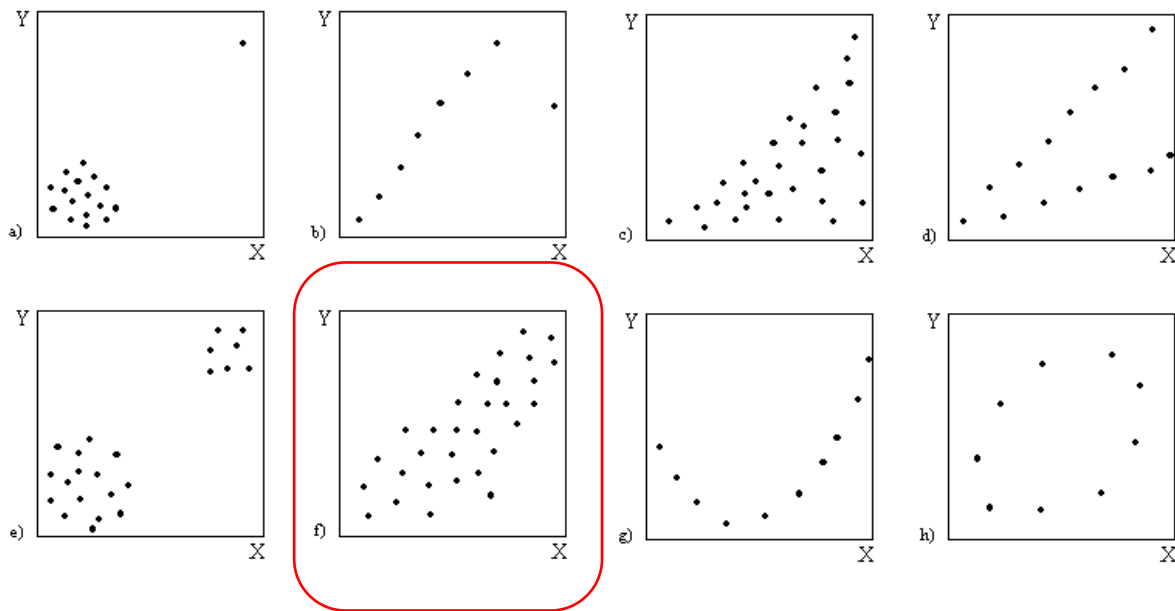
Mistake #1: Excessive Focus on r^2

All of these datasets all have $r^2 = 0.7$



Mistake #1: Excessive Focus on r^2

All of these datasets all have $r^2 = 0.7$



Only this one indicates a valid use of the equation.



Mistake #1: Excessive Focus on r^2

One can often claim a “good” equation based on the r^2 value. The previous graphs reveal how important it is to graph the data.

When the graphs show random variation only:

- $r^2 > 0.4$: An association exists, but other causes also exist
- $r^2 > 0.7$: The relationship can be used with caution
- $r^2 > 0.9$: A useful relationship exists
- $r^2 > 0.95$: The relationship is good

These “rules of thumb” are a subjective assessment based on experience.

They are restricted to the population from which the sample was taken.

Mistake #1: Excessive Focus on r^2

A better indicator than r^2 is to instead consider s , the standard error.

The standard error is the average error about the line.

It is in units of response, and so often easily interpreted.

The prediction is roughly $\pm 2s$ in confidence.

Regression Statistics	
Multiple R	0.997
R Square	0.994
Adjusted R Square	0.993
Standard Error	18.6
Observations	10

r^2

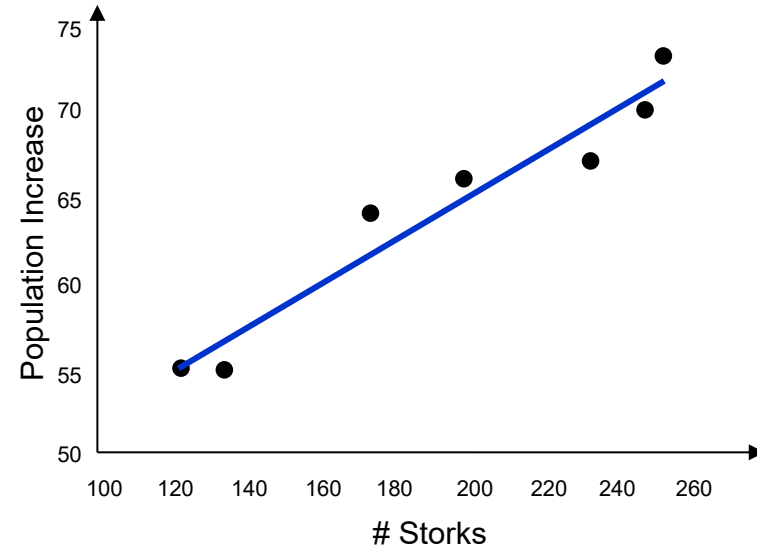
r^2_{adjusted}

Standard error s

Also always use r^2_{adj} not r^2 as a diagnostic.

Mistake #2: Attributing Cause & Effect

*Doubling the number of storks
causes an increase in the population
of Oldenburg by 10,000!*

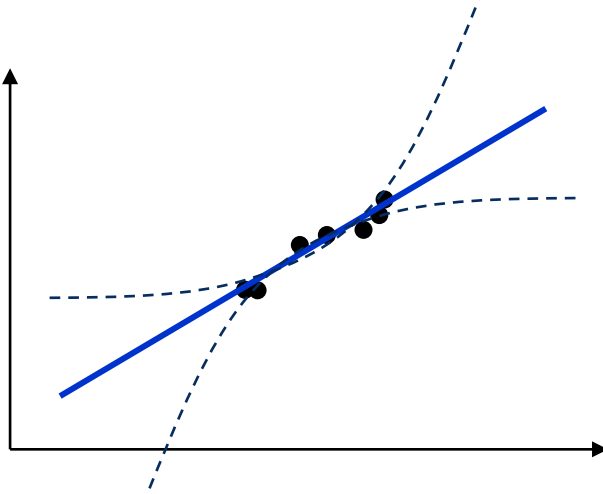


Neyman, J. (1952) Lectures and Conferences on Mathematical Statistics and Probability, 2nd edn, pp. 143-154. Washington DC: US Department of Agriculture.

Correlation does not mean cause-and-effect, only that a relationship exists between the two variables. Something else can cause both.

Mistake #3: Extrapolating

The relationship within the data range need not apply outside the range.

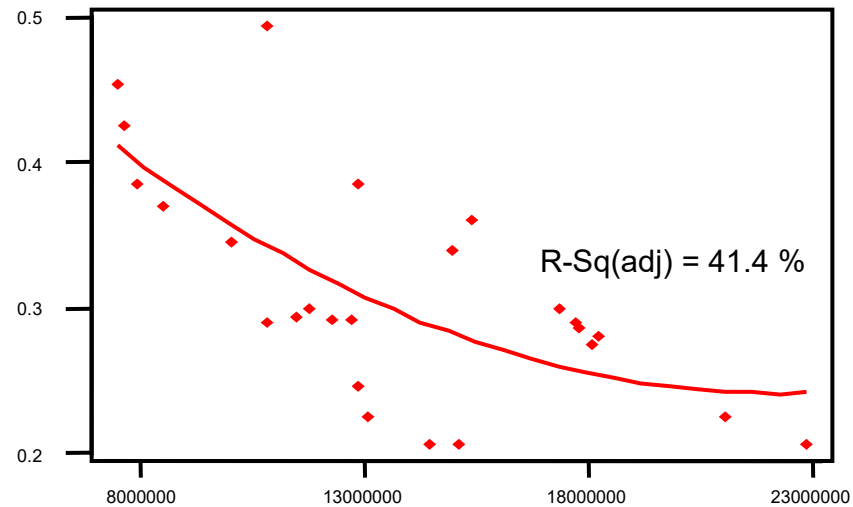


In 176 years, the Lower Mississippi has shortened itself 242 miles. This equates to an average of -1.375 miles/year. Therefore, any calm person, who is not blind or idiotic, can see that in the Old Oolitic Silurian Period, just a million years ago, this river was upward of one million, three hundred thousand miles long, and stuck out over the Gulf of Mexico like a fishing rod. By the same token, any person can see that 742 years from now, the lower Mississippi will be only 1.75 miles long, and Cairo (Illinois) and New Orleans will have joined their streets together.

Mark Twain, 1874

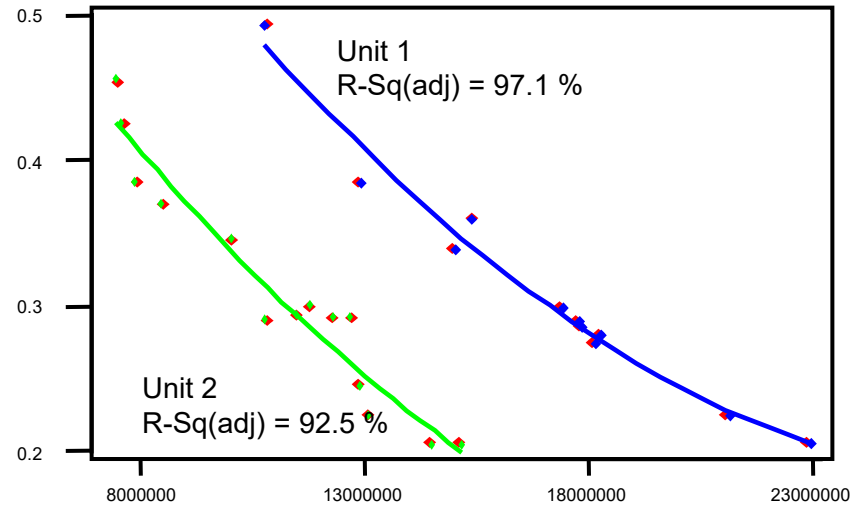
Mistake #4: Misinterpreting the Data

Masking real correlations or creating false correlations



Mistake #4: Misinterpreting the Data

Masking real correlations or creating false correlations

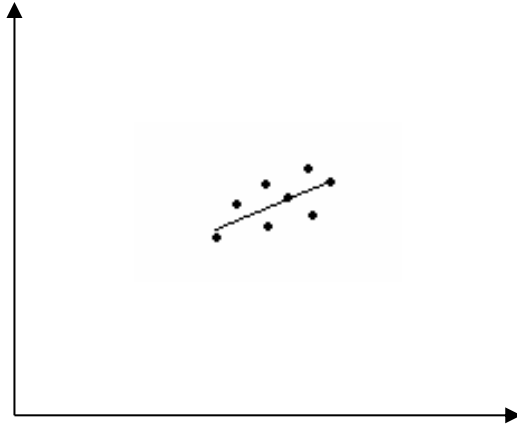


The data is actually a combination from different units.

Mistake #5: Data Over Too Small a Range

A broader range of X will yield a better estimate of the regression line.

Collecting data over too narrow a range

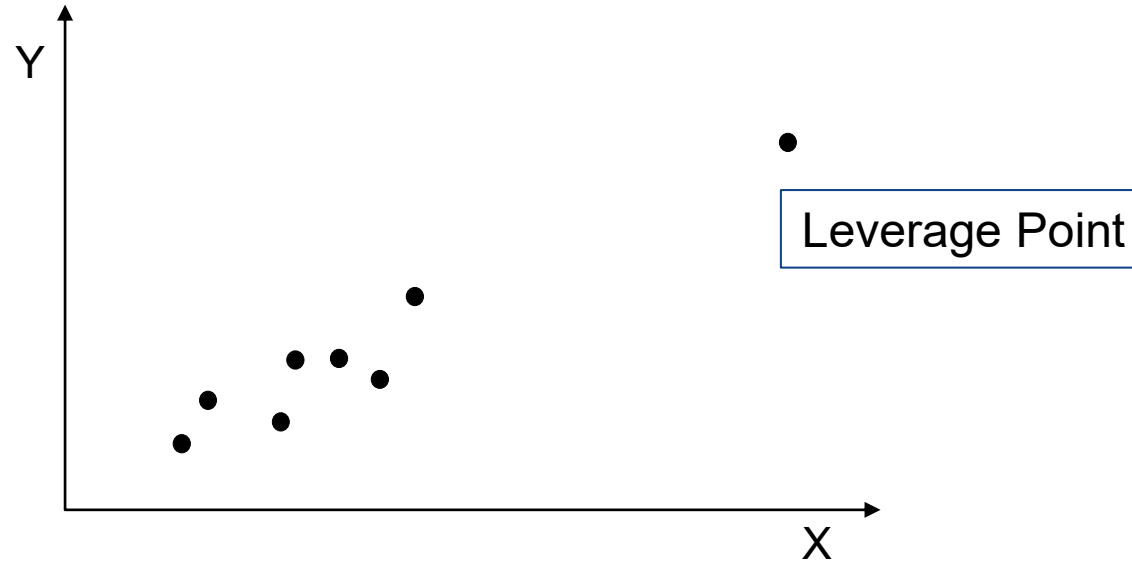


A better estimate through a wider spread

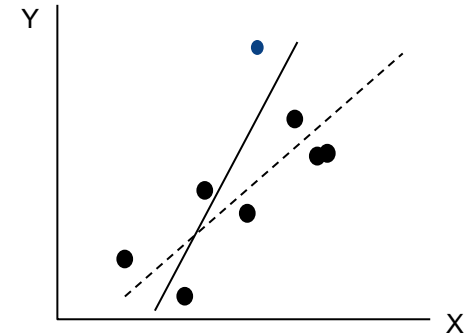
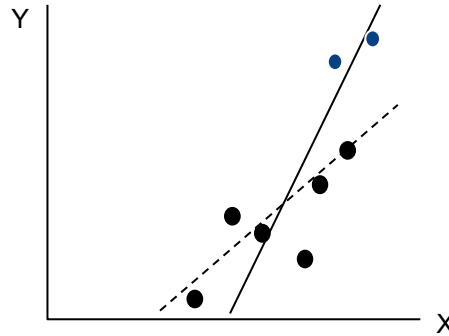
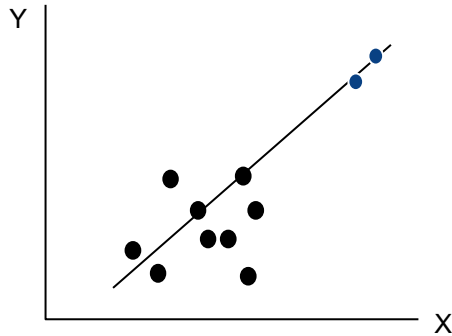
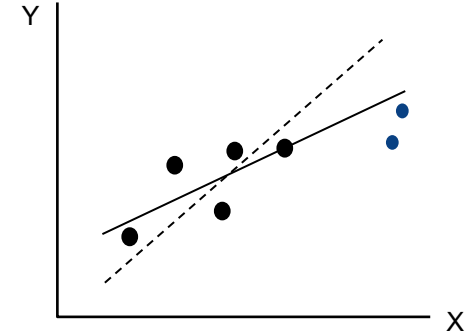
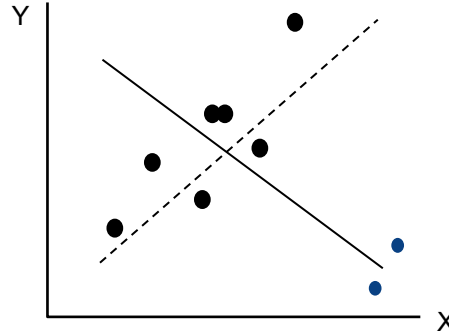
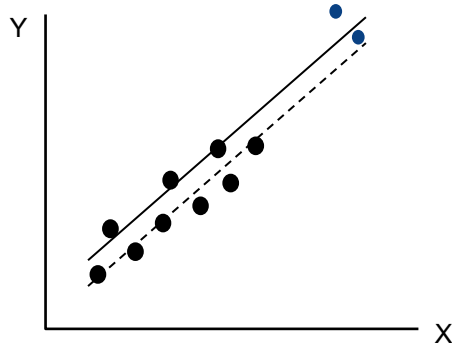


Influential Observations: Leverage Points

A **Leverage Point** is an observation that is extreme in the X -direction and accounts for a high proportion of the sum of squares.

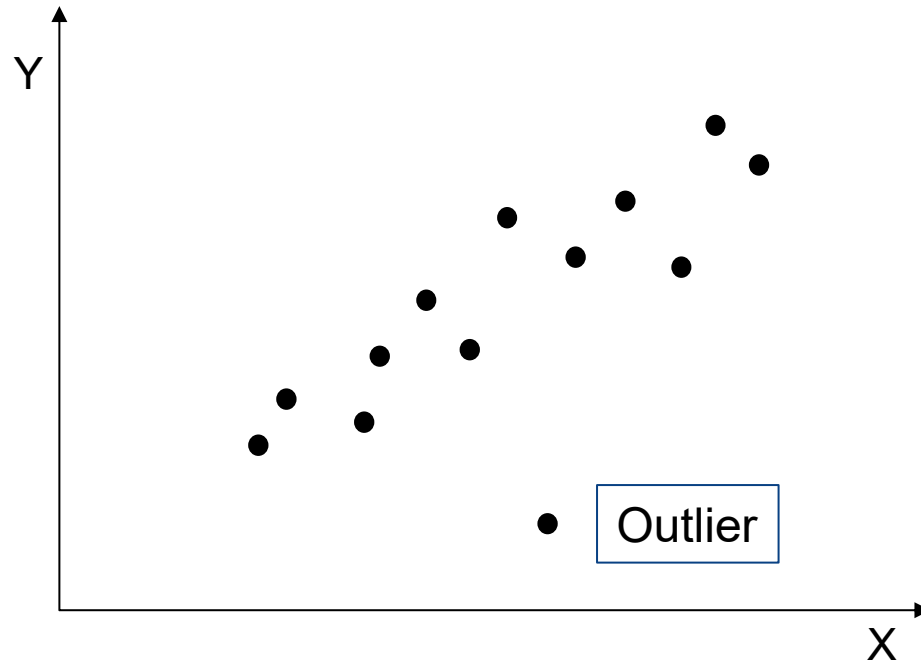


Influential Observations: Leverage Points



Influential Observations : Outliers

An ***outlier*** is a measurement that has a large residual.





Treating Influential Observations

Outlier points can be removed for two reasons:

- They have a reasonable explanation for not being included in the analysis in the first place (e.g. dropped part, input error)
- They invalidate any reasonable interpretation of the regression if they were included

Points removed for reason #2 MUST be reported along with the regression results, either as a count or as a percentage called “bogus” values.

Such points are often a source of lowered performance, and as such, they must not be “hidden.”



Overview

OLS regression

Regression Table

Residual Analysis

Polynomial Regression

Multiple Regression

Over Fitting

Maximum Likelihood Regression

Regression Diagnostics

The objective of developing a regression equation is to be able to perform prediction. However, the user needs to verify if the regression line is meaningful prior to using it for prediction.

As with ANOVA, the regression can be studied with a ***Residual Analysis***





Residual Analysis

Regression Model:

$$Y = \hat{Y} + e$$

where

- Y is the observed response
- $\hat{Y}$ is the predicted response $= mX + b$
- and e is the error or residual, assumed to be:
 - Not significantly not-normal
 - Mean of zero
 - Variance of σ^2
 - Identical distribution over the range of X

Hence, there is a need to verify that the residuals are ***Normally & Identically Distributed*** about the mean = 0 with variance = σ^2



Residual Plots

Histogram of Residuals

Normal Plot of Residuals

Residuals vs Fitted Value

Residuals vs Run Order

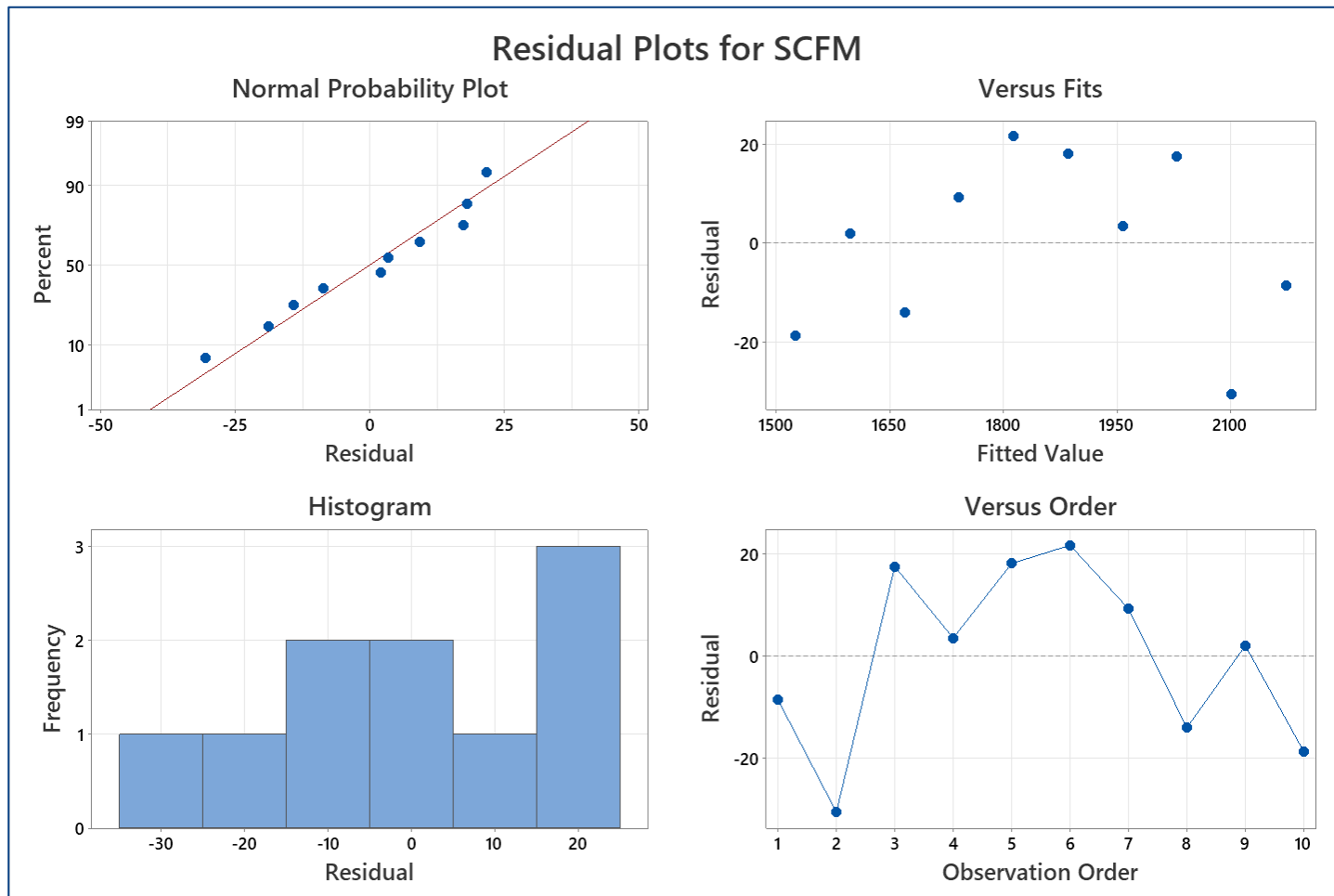
Residuals vs Predictor and Other Variables



Example – Residual Analysis

Perform a residual analysis on the linear regression model obtained for SCFM versus ESP.

Residual Analysis





Overview

OLS regression

Regression Table

Residual Analysis

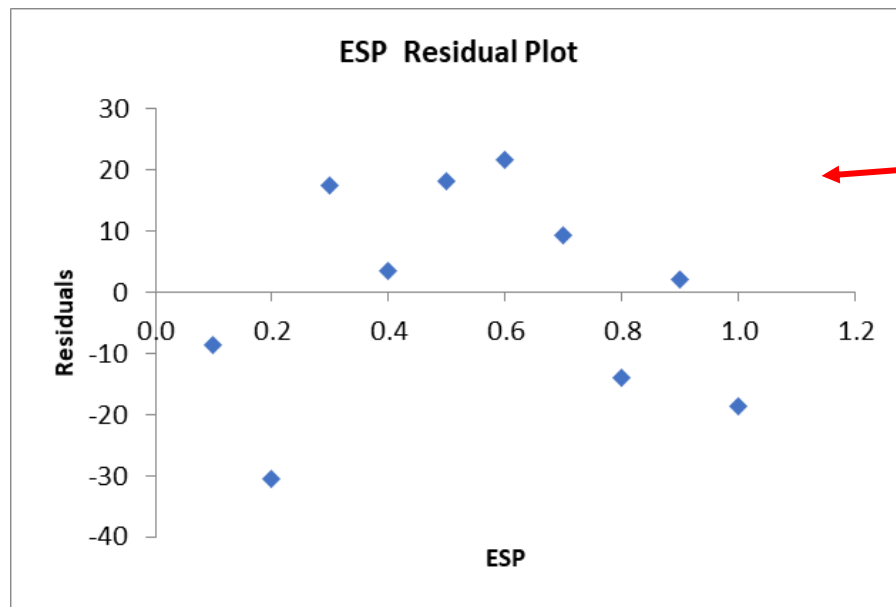
Polynomial Regression

Multiple Regression

Over Fitting

Maximum Likelihood Regression

Polynomial Regression



Curvature in the Residuals vs
Fitted Value Plot?



Polynomial Regression

The linear model fit quite well, but the residual analysis suggests that there could also be significant quadratic terms.

We can seek to improve r^2 by fitting the response using a higher order model (e.g., quadratic, cubed, etc.) of the predictor. That is, the model could be :

$$Y = b_0 + b_1X + b_2X^2 + b_3X^3 + \dots$$

The Polynomial Regression Model is a variant of the simple linear regression model, with higher orders of the predictor.

Example – Polynomial Regression

Can we get a better fit in the model?

Try Quadratic, then try Cubic.

ESP	ESP2	ESP3	SCFM
0.1	0.01	0.001	2165.0
0.2	0.04	0.008	2071.0
0.3	0.09	0.027	2047.1
0.4	0.16	0.064	1961.1
0.5	0.25	0.125	1903.7
0.6	0.36	0.216	1835.3
0.7	0.49	0.343	1750.9
0.8	0.64	0.512	1655.6
0.9	0.81	0.729	1599.7
1.0	1	1	1507.0

Insert two columns, and add entries of
 $ESP \cdot ESP$ and ESP^3



A Better Fit with Polynomial Regression?

Fitting a response versus higher orders of a predictor will invariably improve the coefficient of determination r^2 .

However, this is at the expense of a loss in degrees of freedom.

To compare if a higher order model offers a better fit, two other comparisons must be made:

- Adjusted Coefficient of Determination r_{adjusted}^2
- Standard Error s

Recall the Linear Fit

Using the regression option, create a scatter plot and regression model for SCFM vs ESP

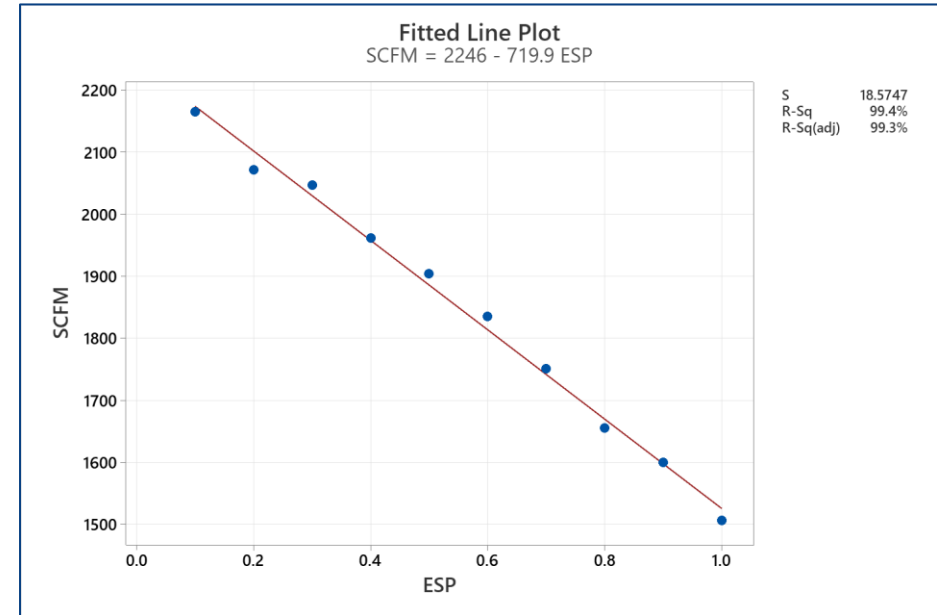
Stat > Regression > Fitted Line Plot...

Model Summary

S	R-sq	R-sq(adj)
18.5747	99.36%	99.28%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	427507	427507	1239.09	0.000
Error	8	2760		345	
Total	9	430267			



Example – Quadratic Fit

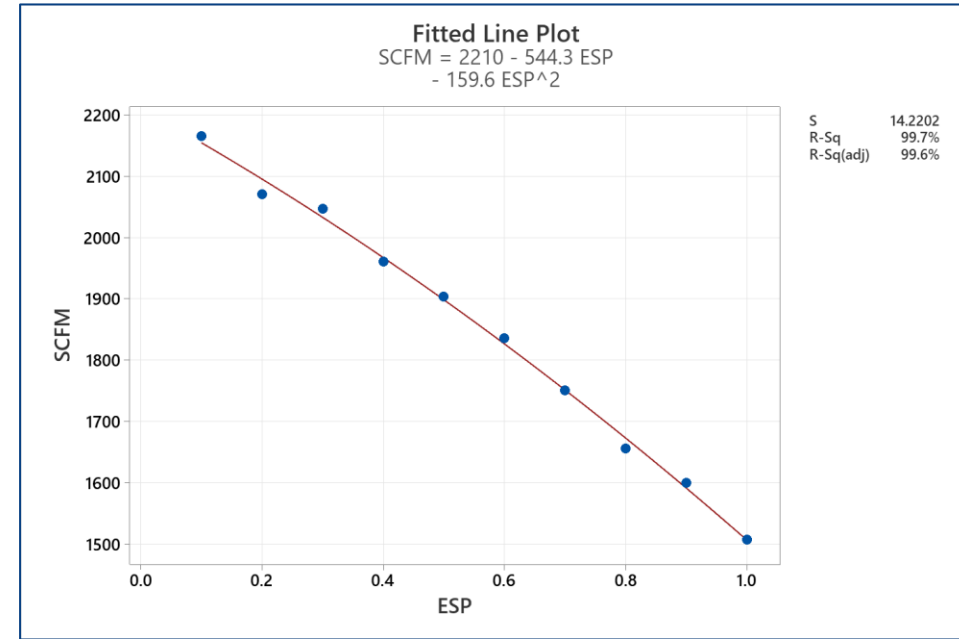
Better!

Model Summary

S	R-sq	R-sq(adj)
14.2202	99.67%	99.58%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	428852	214426	1060.39	0.000
Error	7	1415	202		
Total	9	430267			



Example – Cubic Fit

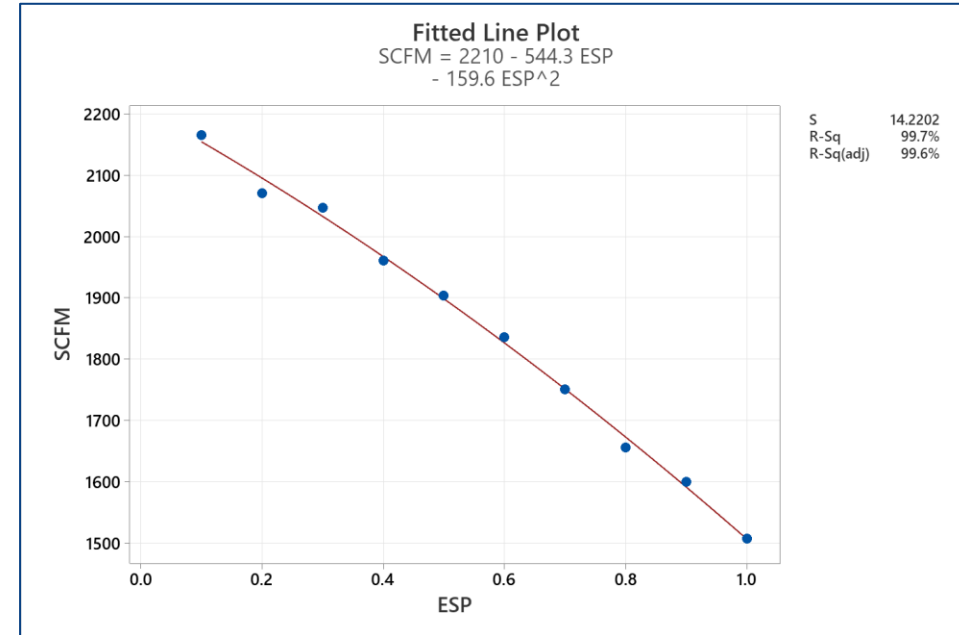
Worse!

Model Summary

S	R-sq	R-sq(adj)
15.3596	99.67%	99.51%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	3	428852	142951	605.94	0.000
Error	6	1415	236		
Total	9	430267			





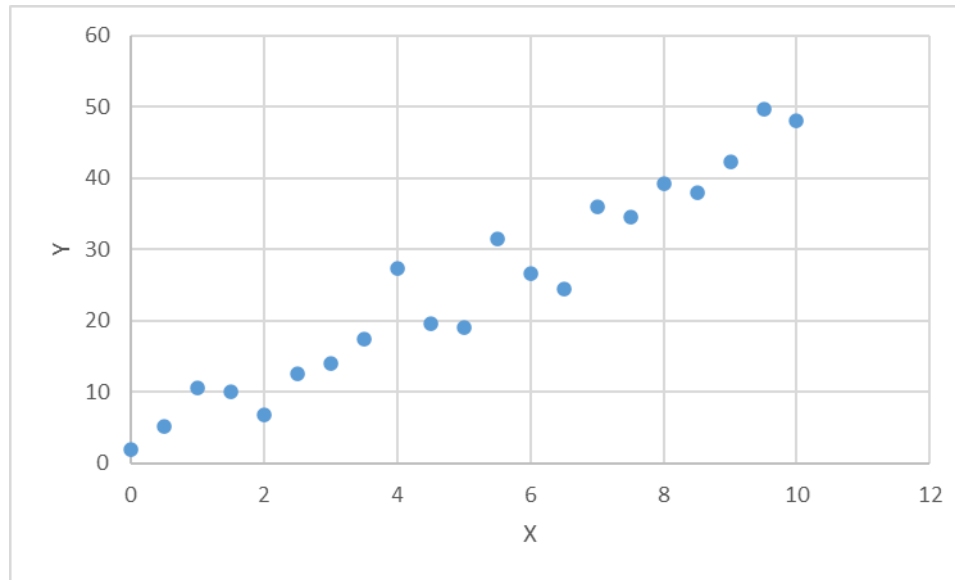
Regression Diagnostics

Don't be infatuated with r^2 . It always gets better with more terms.

Better diagnostics are r_{adj}^2 and the residual error s .

Advanced: Regression and the Constant Term

An advanced topic is to consider the case when the data goes through the origin.
Then the constant term will be insignificantly different from zero.



Advanced: Regression and the Constant Term

Here a better regression is without the insignificant constant term. Drop it.

Notice while r_{adj}^2 got better, s got worse. But it is a better model.

Regression Statistics					
Multiple R	0.971				
R Square	0.942				
Adj R Square	0.939				
Standard Error	3.56				
Observations	21				
ANOVA					
	df	SS	MS	F	P-value
Regression	1	3901.9	3901.9	308.7	0.000
Residual	19	240.2	12.6		
Total	20	4142.1			
	Coeff	Std Error	t Stat	P-value	
Intercept	1.995	1.498	1.33	0.199	
X	4.502	0.256	17.57	0.000	

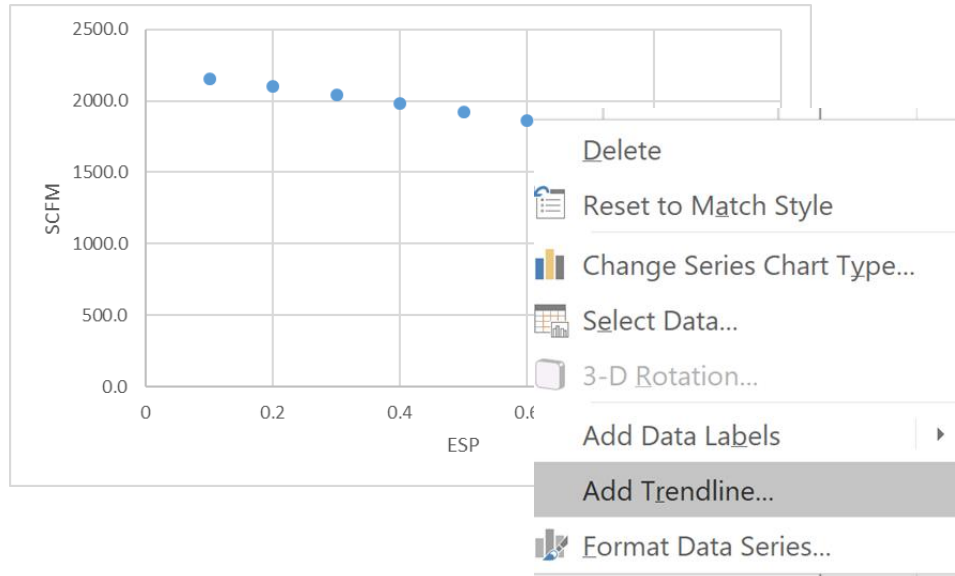
$$y = mx + b$$

Regression Statistics					
Multiple R	0.992				
R Square	0.984				
Adj R Square	0.984				
Standard Error	3.72				
Observations	21				
ANOVA					
	df	SS	MS	F	p-value
Regression	1	16490.9	16490.9	1193.1	0.000
Residual	19	262.6	13.8		
Total	20	16753.5			
	Coeff	std error	t Stat	P-value	
Intercept	0				
X	4.794	0.135	35.4	0.000	

$$y = mx$$

Excel Regression

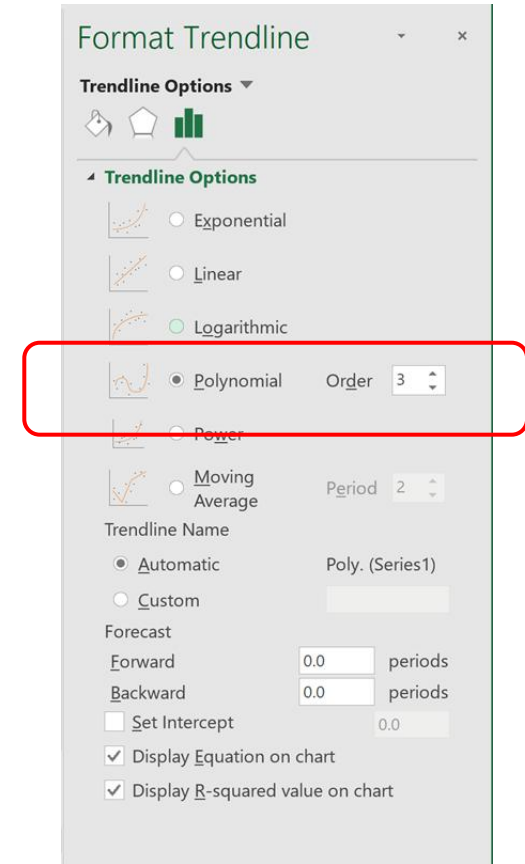
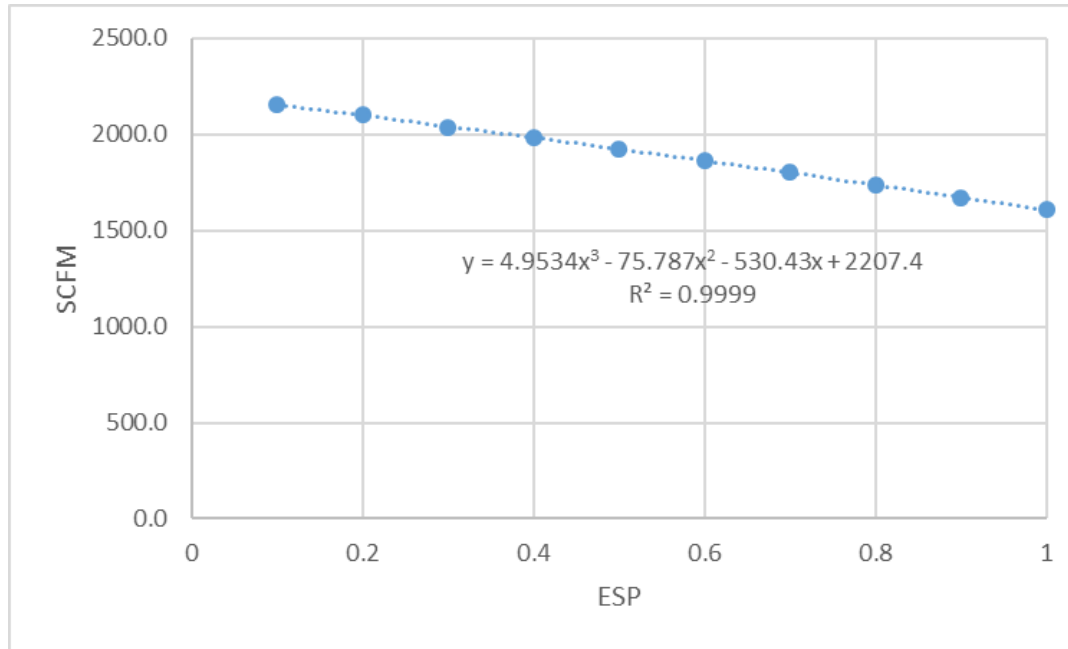
Excel will also provide a fitted line on an x-y graph



Right click on the data series
Select 'Add Trendline'

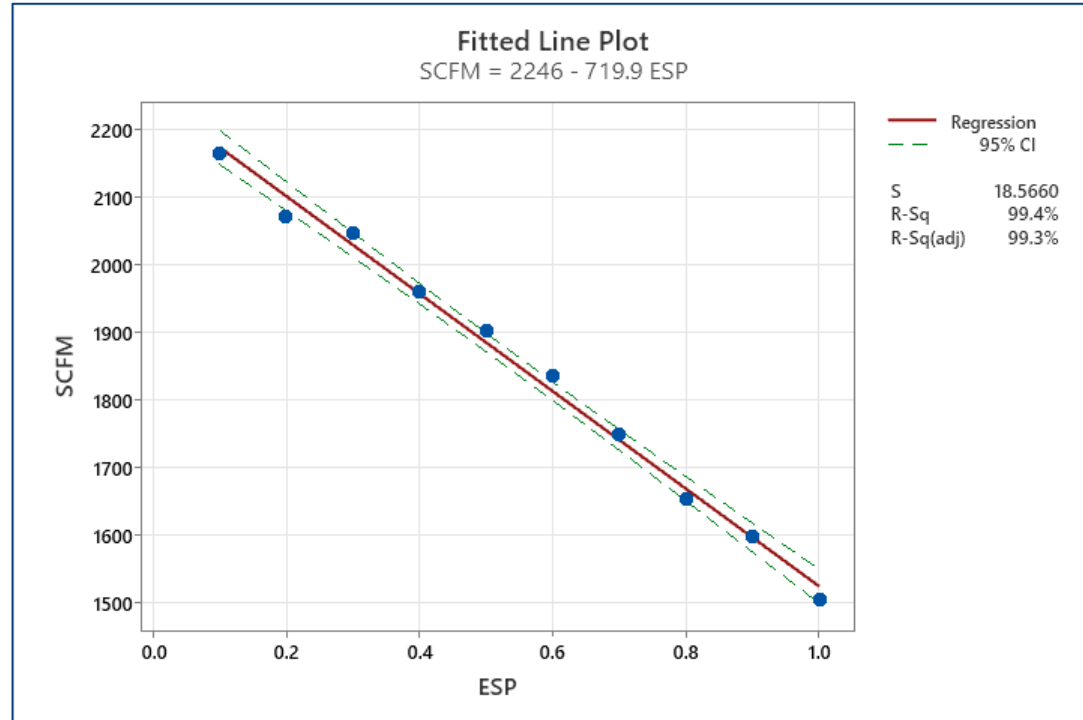
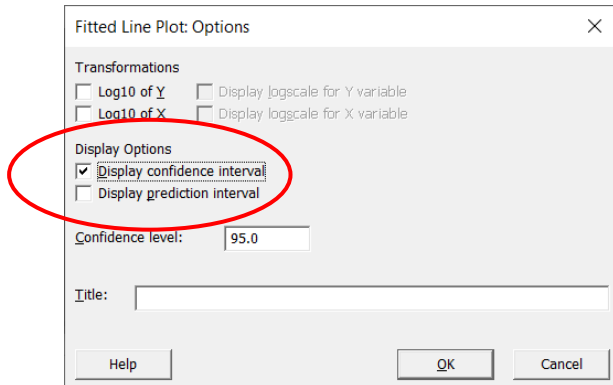
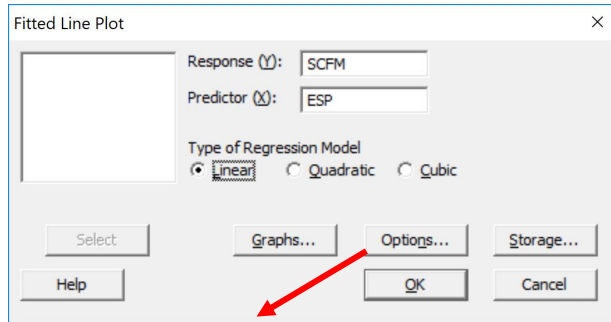
Excel Cubic Regression

And higher order polynomials, but no s or r_{adj}^2 !



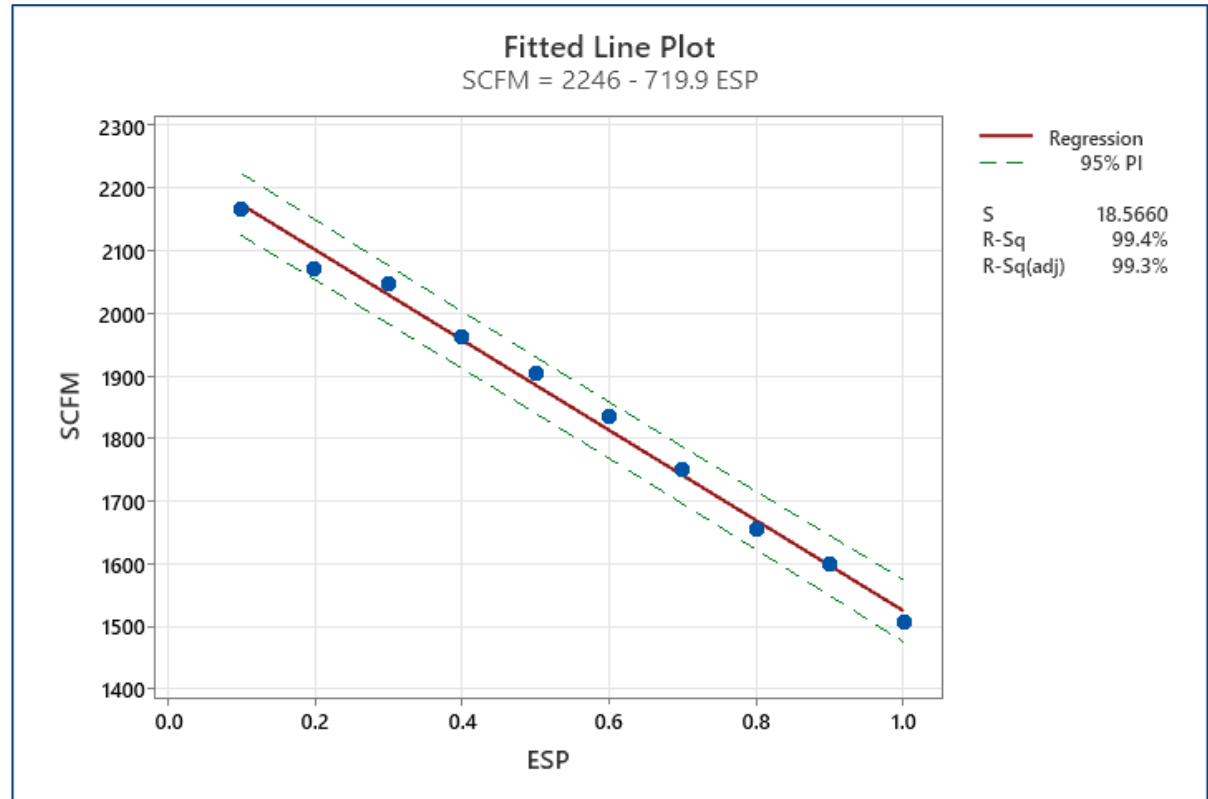
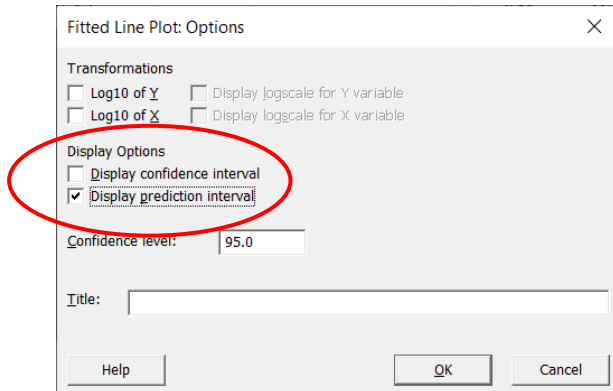
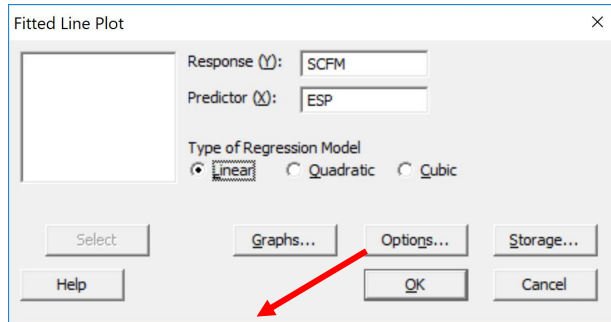
Regression Confidence Bands

A regression **Confidence Interval** places confidence interval bounds on for the Y fit equation.



Regression Confidence Bands

A regression *Prediction Interval* places confidence interval bounds for the Y individual values.



Polynomial Regression in Minitab

Linear Fit

Fitted Line Plot

C1 ESP
C2 SCFM

Response (Y): SCFM

Predictor (X): ESP

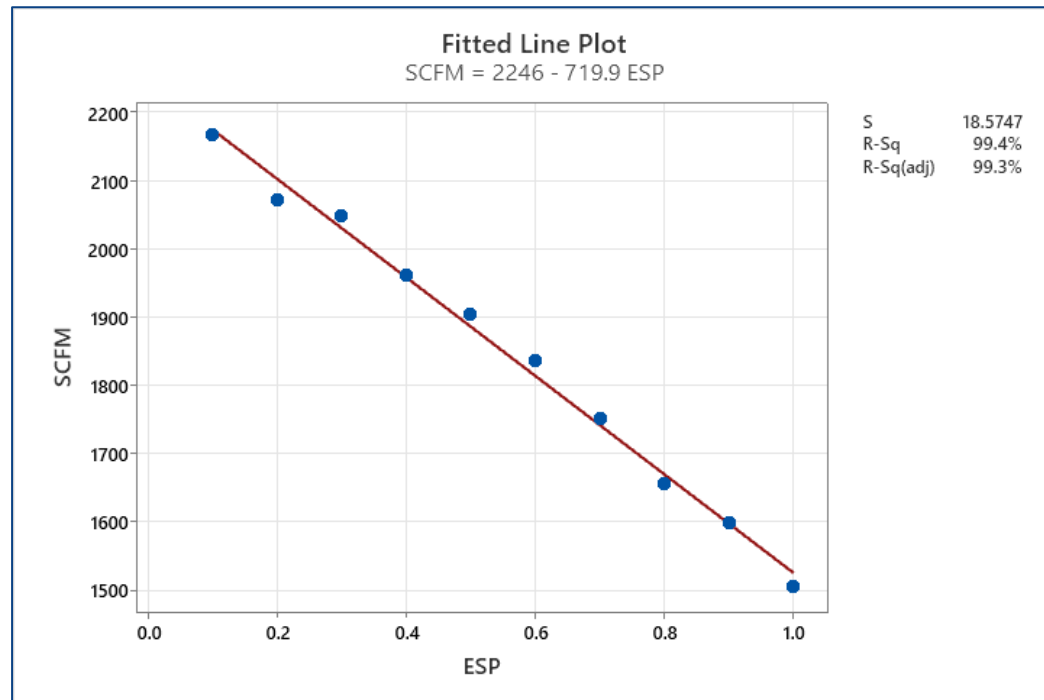
Type of Regression Model
☒ Linear ☐ Quadratic ☐ Cubic

Select Graphs... Options... Storage...

Help OK Cancel

Model Summary

S	R-sq	R-sq(adj)
18.5747	99.3585%	99.2783%



Polynomial Regression in Minitab

Quadratic fit

Fitted Line Plot

Response (Y): SCFM

Predictor (X): ESP

Type of Regression Model

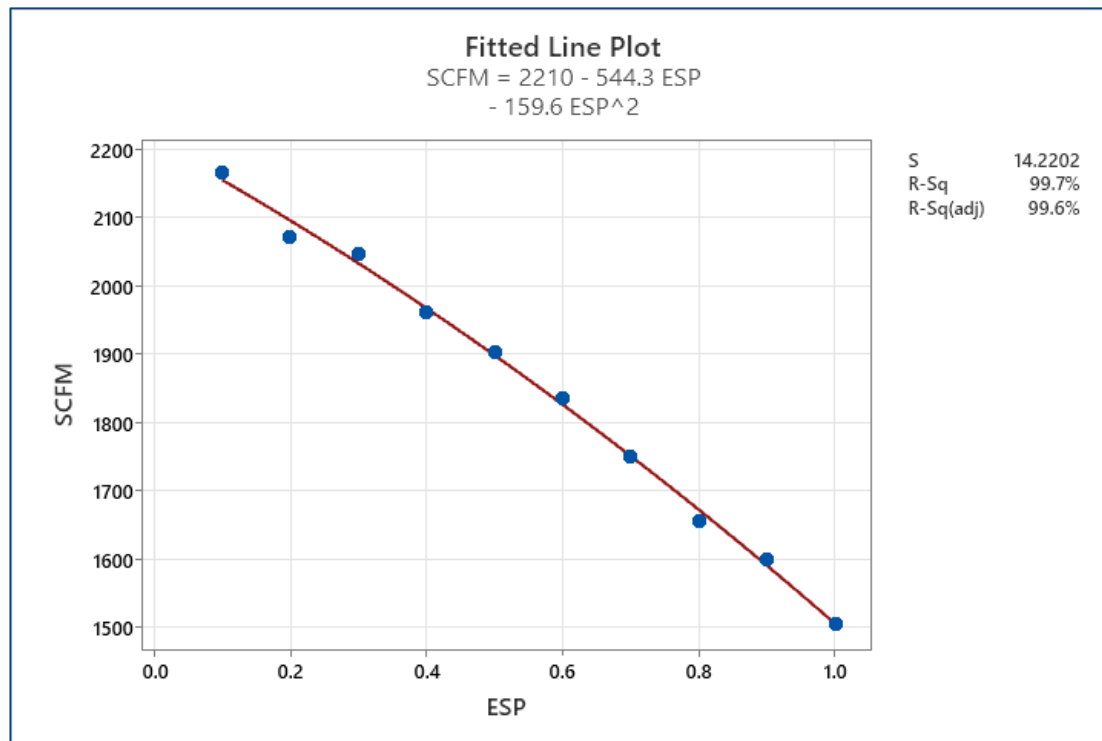
☐ Linear ☒ Quadratic ☐ Cubic

Select Graphs... Options... Storage...

Help OK Cancel

Model Summary

S	R-sq	R-sq(adj)
14.2202	99.6710%	99.5770%



Polynomial Regression in Minitab

Cubic fit

Fitted Line Plot

Response (Y): SCFM

Predictor (X): ESP

Type of Regression Model

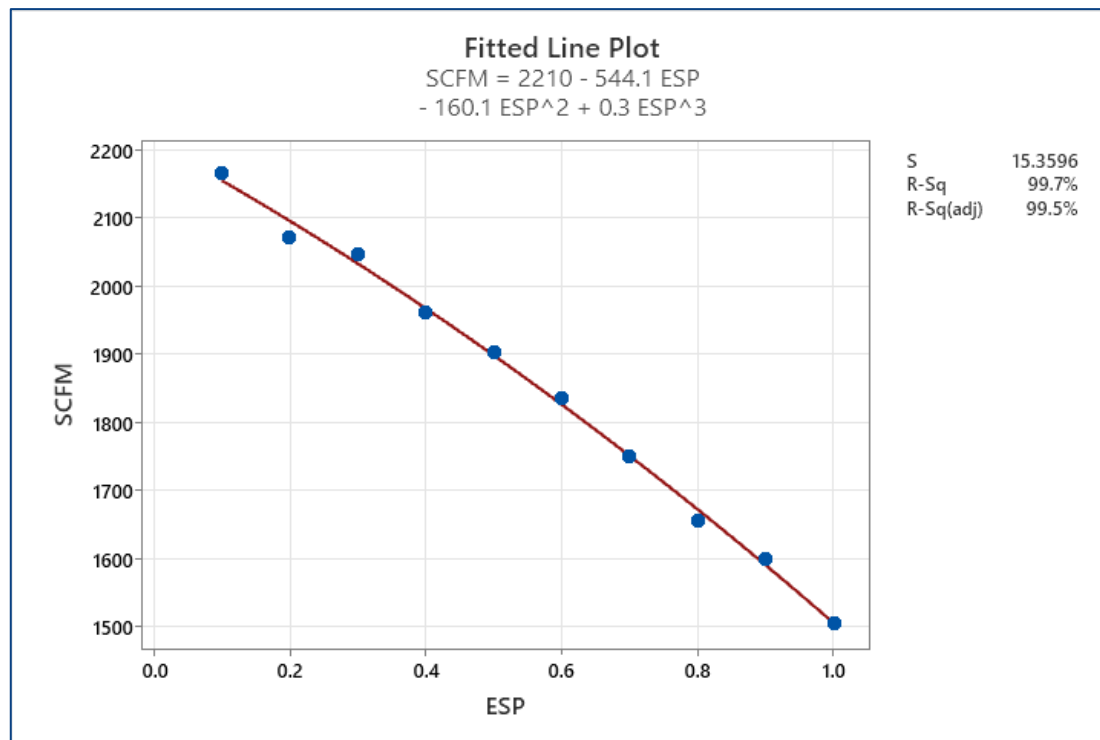
☐ Linear ☐ Quadratic ☒ Cubic

Select Graphs... Options... Storage...

Help OK Cancel

Model Summary

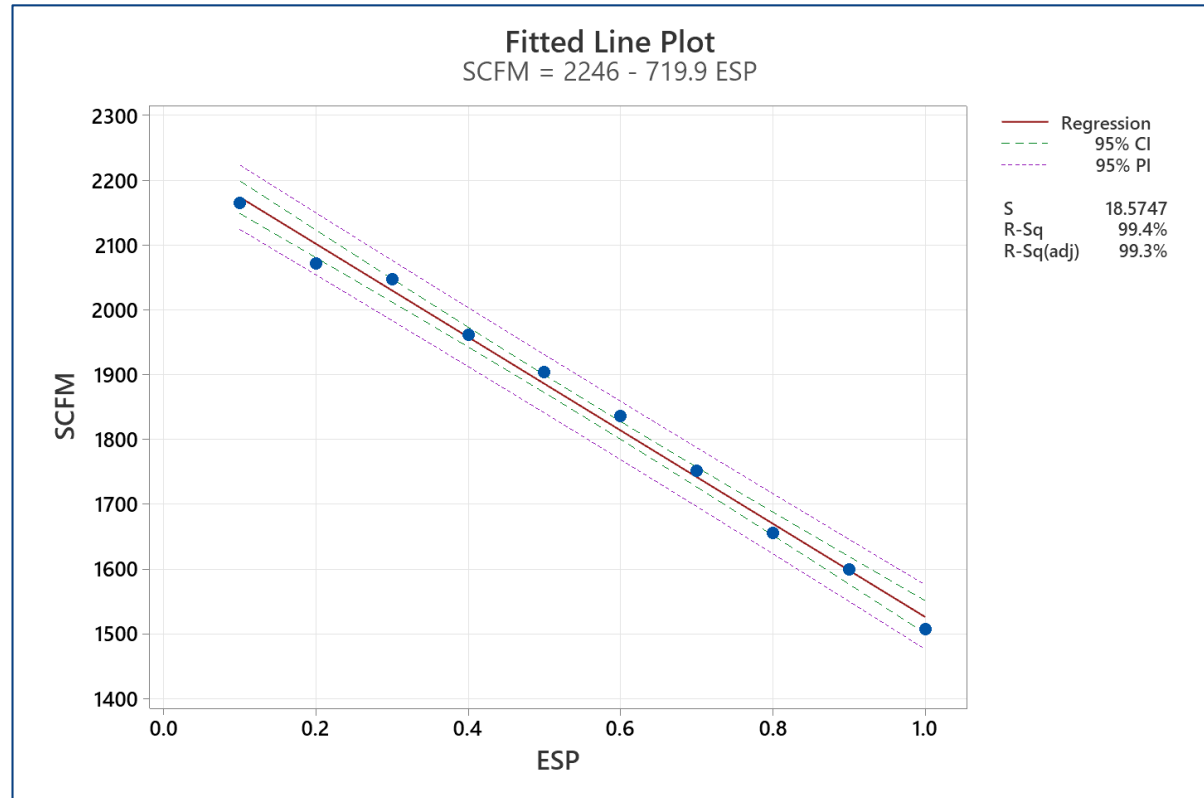
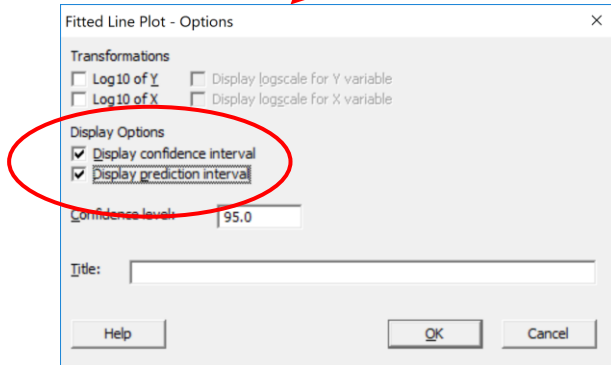
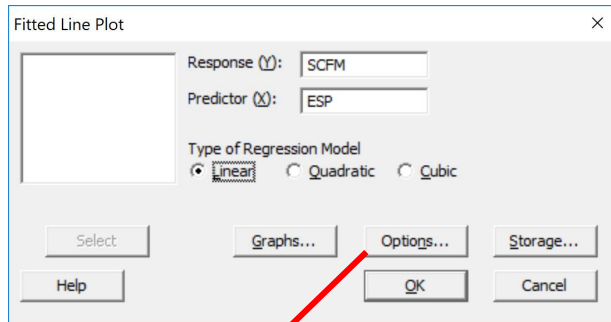
S	R-sq	R-sq(adj)
15.3596	99.6710%	99.5065%



Regression Confidence Bands

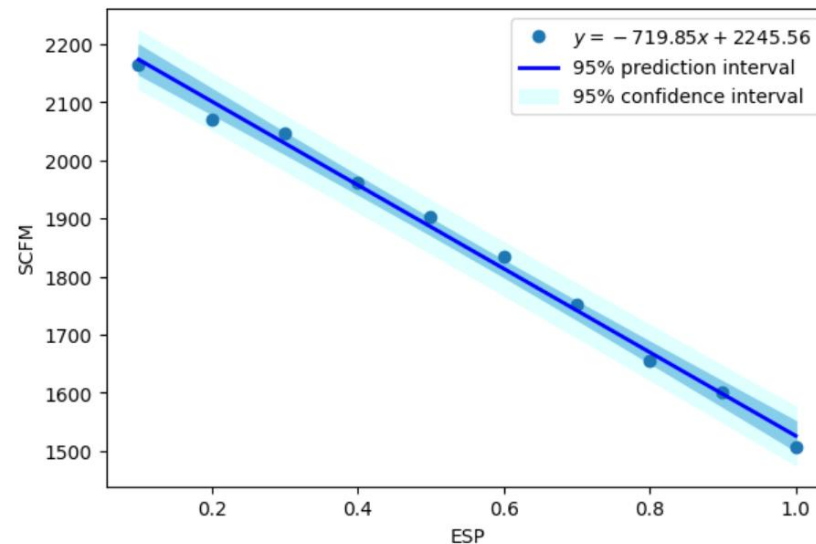
A regression **Confidence Interval** is for the Y fit equation at a given X value.

A regression **Prediction Interval** is for the Y individual values at a given X value.



Linear Regression – Python

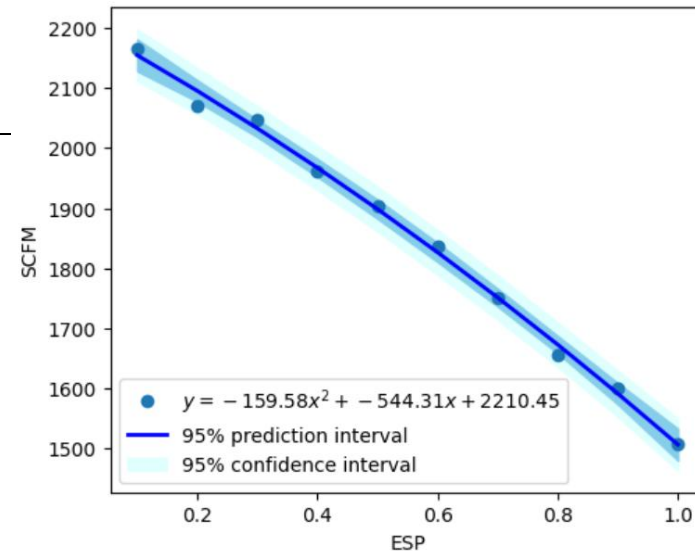
```
lreg = ols('SCFM ~ ESP', data=df).fit()
st, data, ss2 = summary_table(lreg, alpha=0.05)
fittedvalues = data[:, 2]
predict_mean_se = data[:, 3]
predict_mean_ci_low, predict_mean_ci_upp = data[:, 4:6].T
predict_ci_low, predict_ci_upp = data[:, 6:8].T
x = df['ESP']
y = df['SCFM']
with Figure(6, 4) as (fig, ax):
    ax.plot(x, y, 'o')
    ax.plot(x, fittedvalues, 'b-', lw=2)
    ax.fill_between(x, predict_ci_upp, predict_ci_low,
                    color = 'lightcyan')
    ax.fill_between(x, predict_mean_ci_upp, predict_mean_ci_low,
                    color = 'skyblue')
    ax.set_xlabel("ESP")
    ax.set_ylabel("SCFM")
    ax.legend([
        f'$y = {slope:.2f}x + {intercept:.2f}$',
        '95% prediction interval',
        '95% confidence interval'
    ])
])
```



Quadratic Regression – Python

```
qreg = ols('SCFM ~ ESP + ESP2', data=qfit).fit()
st, data, ss2 = summary_table(qreg, alpha=0.05)
fittedvalues = data[:, 2]
predict_mean_se = data[:, 3]
predict_mean_ci_low, predict_mean_ci_upp = data[:, 4:6].T
predict_ci_low, predict_ci_upp = data[:, 6:8].T

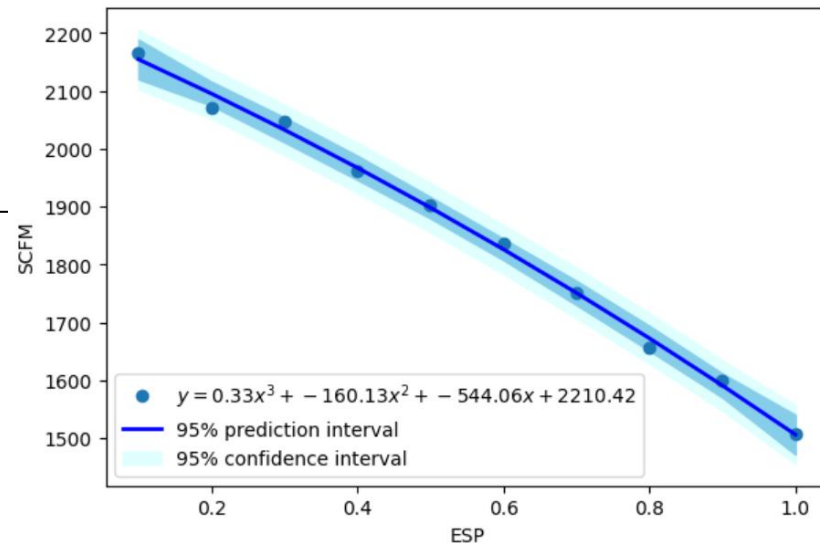
x = df['ESP']
y = df['SCFM']
with Figure(5, 4) as (fig, ax):
    ax.plot(x, y, 'o')
    ax.plot(x, fittedvalues, 'b-', lw=2)
    ax.fill_between(x, predict_ci_upp, predict_ci_low, color = 'lightcyan')
    ax.fill_between(x, predict_mean_ci_upp, predict_mean_ci_low, color = 'skyblue')
    ax.set_ylabel("SCFM")
    ax.set_xlabel("ESP")
    ax.legend([
        f'$y={quadratic:.2f}x^2 + {slope:.2f}x + {intercept:.2f}$',
        '95% prediction interval',
        '95% confidence interval',
    ])
])
```



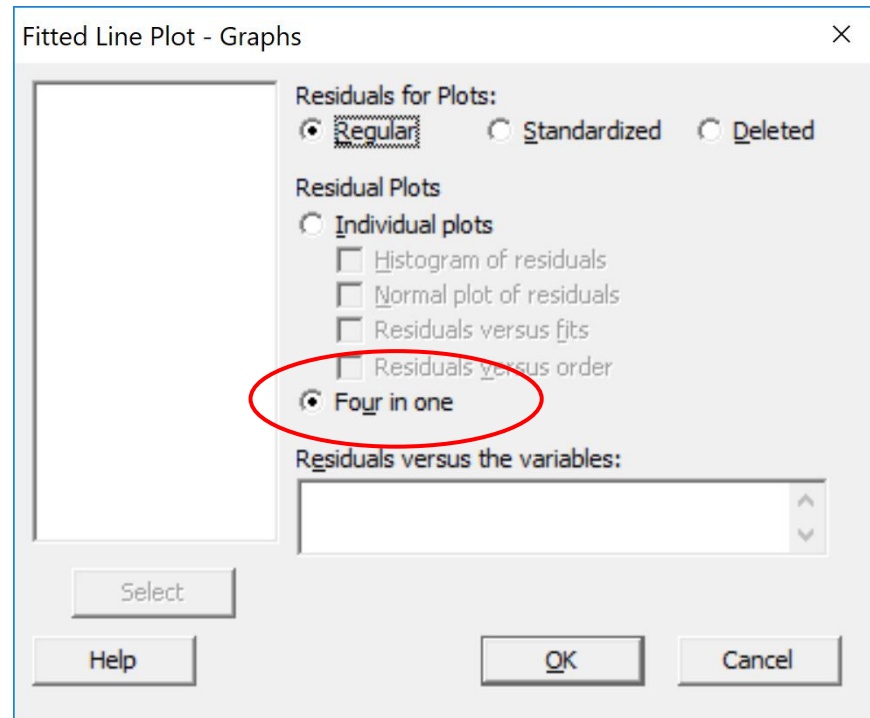
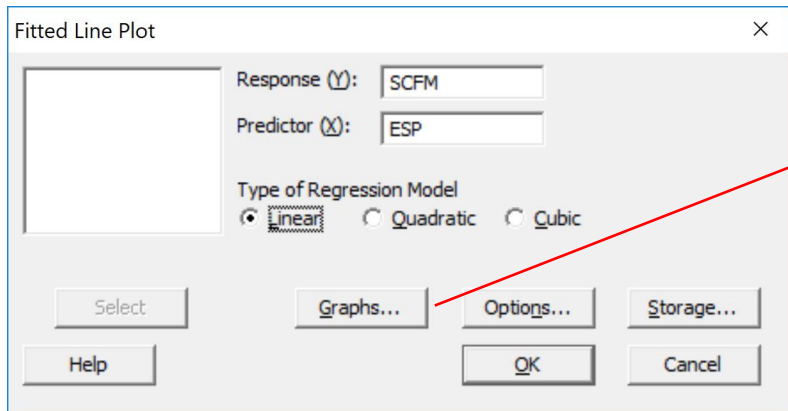
Cubic Regression – Python

```
creg = ols('SCFM ~ ESP + ESP2 + ESP3', data=cfit).fit()
st, data, ss2 = summary_table(creg, alpha=0.05)
fittedvalues = data[:, 2]
predict_mean_se = data[:, 3]
predict_mean_ci_low, predict_mean_ci_upp = data[:, 4:6].T
predict_ci_low, predict_ci_upp = data[:, 6:8].T
```

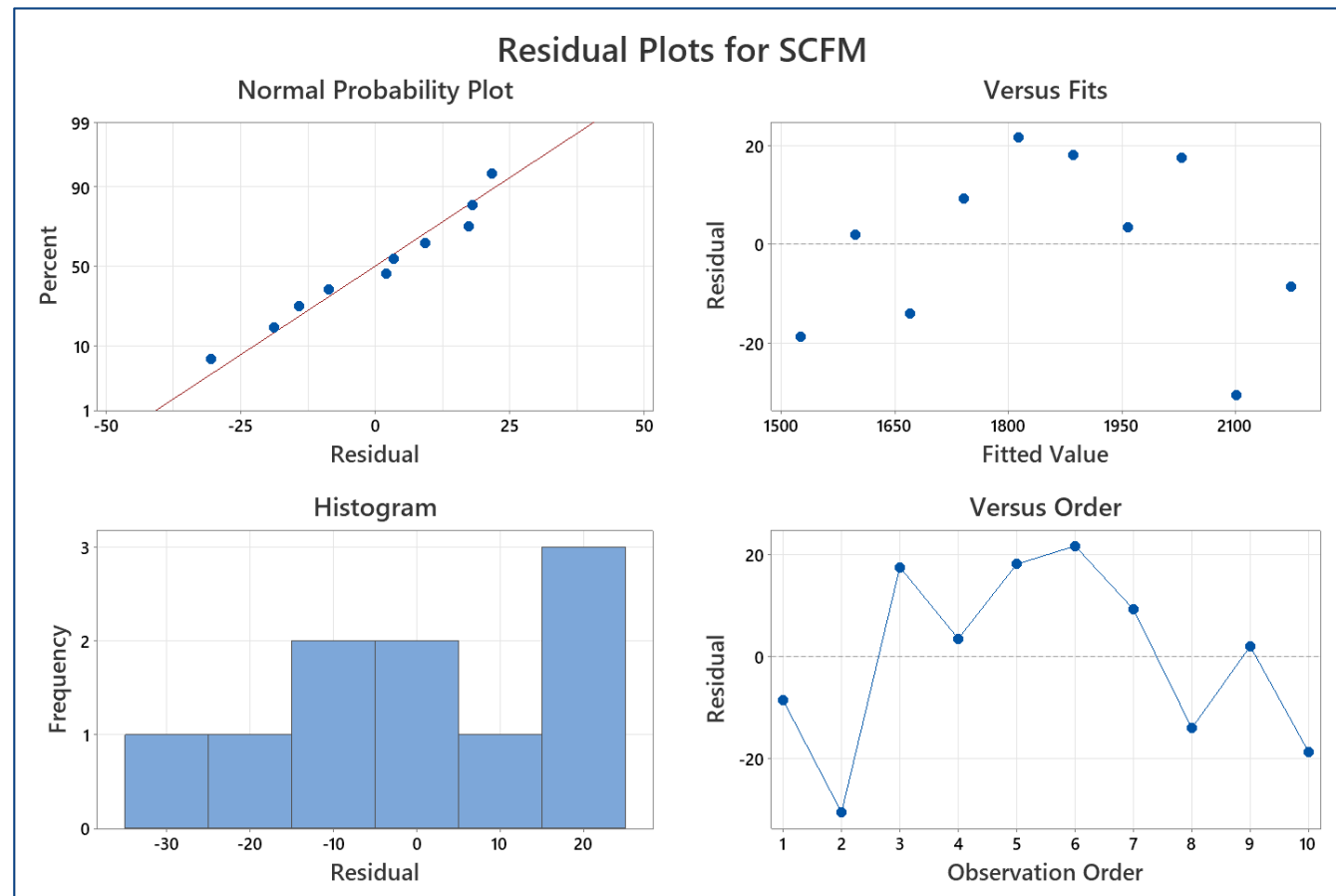
```
x = df['ESP']
y = df['SCFM']
with Figure(6, 4) as (fig, ax):
    ax.plot(x, y, 'o')
    ax.plot(x, fittedvalues, 'b-', lw=2)
    ax.fill_between(x, predict_ci_upp, predict_ci_low, color = 'lightcyan')
    ax.fill_between(x, predict_mean_ci_upp, predict_mean_ci_low, color = 'skyblue')
    ax.set_xlabel("ESP")
    ax.set_ylabel("SCFM")
    ax.legend([
        f'$y = {cubic:.2f}x^3 + {quadratic:.2f}x^2 + {slope:.2f}x + {intercept:.2f}$',
        '95% prediction interval',
        '95% confidence interval'
    ])
})
```



Residual Analysis – Minitab

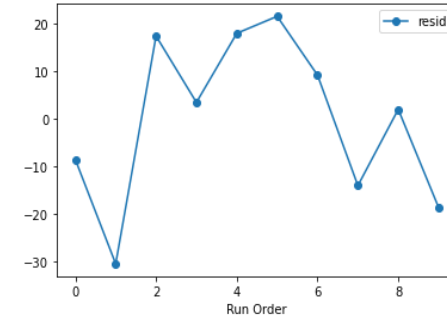
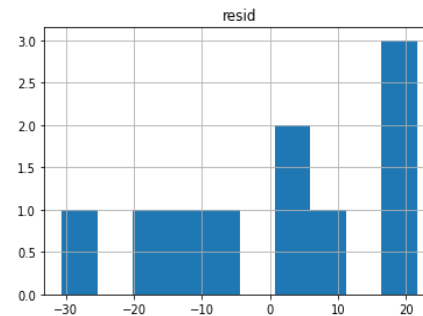
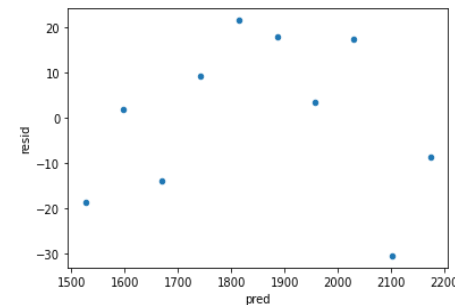
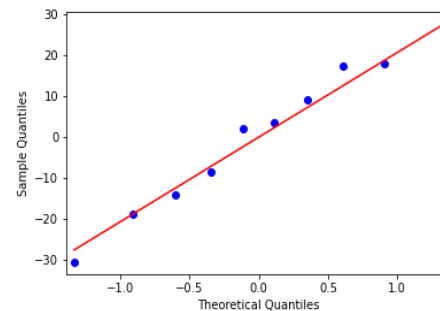


Residual Analysis – Minitab



Residual Analysis – Python

```
lreg = ols('SCFM ~ ESP', data=df).fit()
lfit = df[['ESP', 'SCFM']].copy()
lfit['resid'] = lreg.resid
lfit['pred'] = lfit['SCFM']-lfit['resid']
sm.qqplot(lfit['resid'], line='r')
lfit.plot.scatter(y='resid', x='pred')
lfit.hist(column='resid')
ax = lfit.plot(y='resid', marker='o')
ax.set_xlabel("Run Order")
```





Overview

OLS regression

Regression Table

Residual Analysis

Polynomial Regression

Multiple Regression

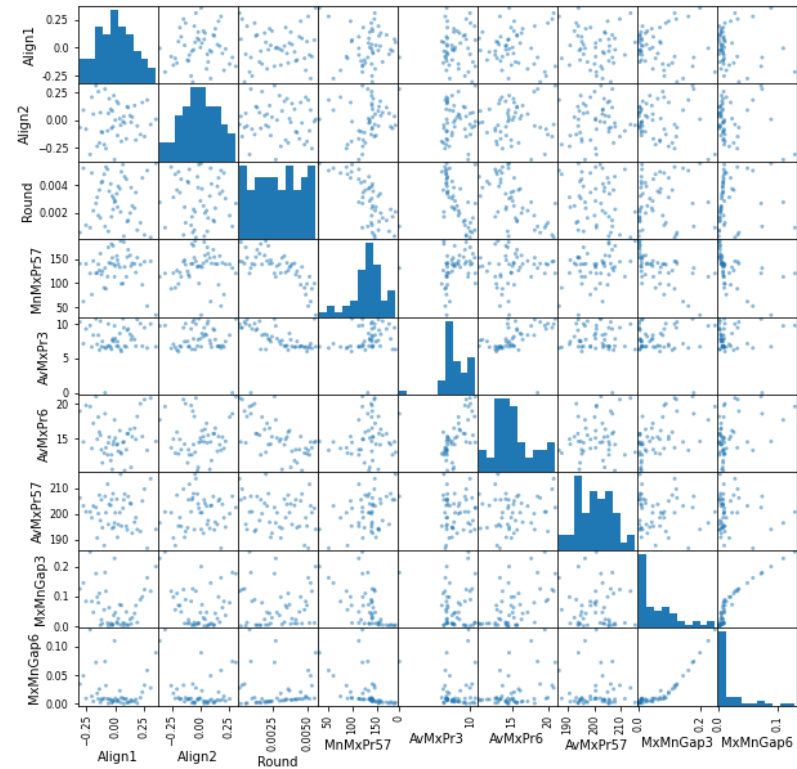
Over Fitting

Maximum Likelihood Regression

Measured Response Fitting

Recall the matrix-plot data from process and inspection measurements.

Can we fit an equation of the inspection results using the process variable measurements?





Multiple Linear Regression

General linear regression model

$$Y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \cdots + b_kx_k$$

where

- Y is the response
- x_i are the predictors
- b_i 's are the regression coefficients



Multiple Linear Regression

Method of determining the statistical relationship between a response variable (Y) and several predictor variables ($X_1, X_2, \dots, X_n$)

$$Y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_kx_k$$

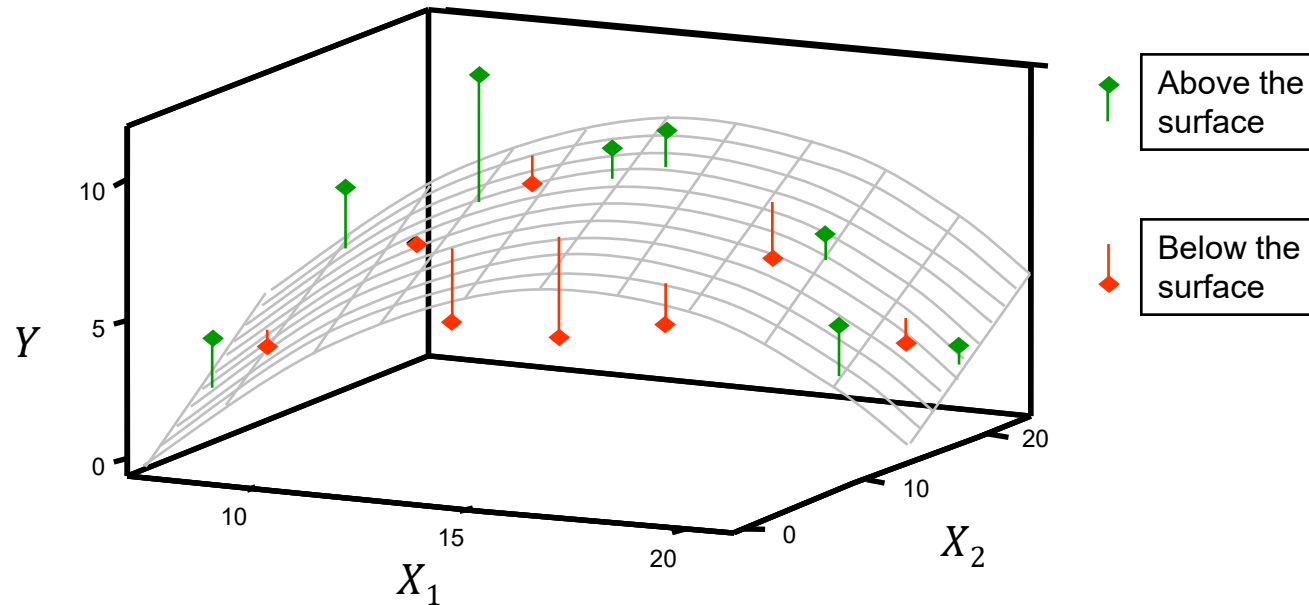
The model is called “linear” because Y is linear in the parameters $b_0, \dots, b_k$, not because Y is a linear function of the X 's.

$$Y = b_0 + b_1x_1 + b_2\sin(x_2) + b_3\cos(x_3)$$

The fitting equation has to be separable into additive terms, each with a b_i coefficient. Otherwise, non-linear regression is needed using an search minimization approach.

Multiple Linear Regression

Ordinary least squares regression attempts to find the best n-dimensional surface that fits the data points (using the method of least squares)



Multiple Regression

Multiple regression is considerably more difficult than simple regression:

- Choice of the best model
- Visualization of the fitted model
- Interpretation of the fitted model
- Computation of the fitted model



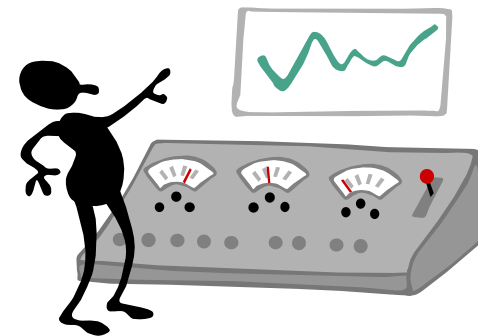
Multiple Regression with Natural Data

Given its complications, multiple regression is very useful in analysis of:

- Naturally occurring experimental research data
- Observational data (passive data)
- Data taken along operating input curves of multi-dimensional inputs

Natural data is more difficult to use than orthogonal DOE data

- Care needs to be taken that the input data reflects a wide range of parameter space
- Observational data often has many of its predictor variables highly correlated



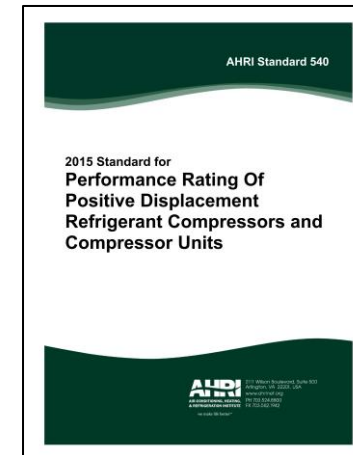
Example – Compressor Map

The global turbomachinery industry is in part self-regulated on practices through the Air-Conditioning, Heating and Refrigeration Institute (AHRI) who define standards and tests.

The AHRI compressor map requires a 10 term third order polynomial fit for Capacity, Power and Mass flow rate in terms of Suction and Discharge Temperatures

$$Y = b_1 + b_2T_s + b_3T_d + b_4T_s^2 + b_5T_sT_d + b_6T_d^2 + b_7T_s^3 + b_8T_s^2T_d + b_9T_sT_d^2 + b_{10}T_d^3$$

These maps are published by suppliers.





Example – Compressor Map

The following capacity data was measured for a compressor.

What are the best map coefficients?

Capacity (BTU/hr)		Discharge Temp (F)					
		90	94	100	106	114	120
Suction Temp (F)	45	40,754	40,274	38,591	37,327	35,442	33,474
	50	45,019	43,801	42,943	40,979	38,484	37,161
	55	49,478	47,984	46,465	44,895	42,827	41,471



Example – Compressor Map

Data must be in matrix form for use by regression tools

Ts (F)	Td (F)	Capacity
45	90	40754
45	94	40274
45	100	38591
45	106	37327
45	114	35442
45	120	33474
50	90	45019
50	94	43801
50	100	42943
50	106	40979
50	114	38484
50	120	37161
55	90	49478
55	94	47984
55	100	46465
55	106	44895
55	114	42827
55	120	41471

Example – Compressor Map

Expand input data to include all quadratic and cubic terms

Ts	Td	Ts2	TsTd	Td2	Ts3	Ts2Td	TsTd2	Td3	Capacity
45	90	2025	4050	8100	91125	182250	364500	729000	40754
45	94	2025	4230	8836	91125	190350	397620	830584	40274
45	100	2025	4500	10000	91125	202500	450000	1000000	38591
45	106	2025	4770	11236	91125	214650	505620	1191016	37327
45	114	2025	5130	12996	91125	230850	584820	1481544	35442
45	120	2025	5400	14400	91125	243000	648000	1728000	33474
50	90	2500	4500	8100	125000	225000	405000	729000	45019
50	94	2500	4700	8836	125000	235000	441800	830584	43801
50	100	2500	5000	10000	125000	250000	500000	1000000	42943
50	106	2500	5300	11236	125000	265000	561800	1191016	40979
50	114	2500	5700	12996	125000	285000	649800	1481544	38484
50	120	2500	6000	14400	125000	300000	720000	1728000	37161
55	90	3025	4950	8100	166375	272250	445500	729000	49478
55	94	3025	5170	8836	166375	284350	485980	830584	47984
55	100	3025	5500	10000	166375	302500	550000	1000000	46465
55	106	3025	5830	11236	166375	320650	617980	1191016	44895
55	114	3025	6270	12996	166375	344850	714780	1481544	42827
55	120	3025	6600	14400	166375	363000	792000	1728000	41471

Example – Compressor Map

Run the regression using all of the AHRI terms

Ts	Td	Ts2	TsTd	Td2	Ts3	Ts2Td	TsTd2	Td3	Capacity
45	90	2025	4050	8100	91125	182250	364500	729000	40754
45	94	2025	4230	8836	91125	190350	397620	830584	40274
10000	91125	202500	450000	1000000					38591
11236	91125	214650	505620	1191016					37327
12996	91125	230850	584820	1481544					35442
14400	91125	243000	648000	1728000					33474
8100	125000	225000	405000	729000					45019
8836	125000	235000	441800	830584					43801
10000	125000	250000	500000	1000000					42943
11236	125000	265000	561800	1191016					40979
12996	125000	285000	649800	1481544					38484
14400	125000	300000	720000	1728000					37161
8100	166375	272250	445500	729000					49478
8836	166375	284350	485980	830584					47984
10000	166375	302500	550000	1000000					46465
11236	166375	320650	617980	1191016					44895
12996	166375	344850	714780	1481544					42827
14400	166375	363000	792000	1728000					41471

Regression ? X

Input

Input Y Range: \$M\$14:\$M\$32 ↑

Input X Range: \$O\$14:\$W\$32 ↑

☒ Labels ☐ Constant is Zero

☐ Confidence Level: 95 %

Output options

☒ Output Range: \$Z\$14 ↑

☐ New Worksheet Ply:

☐ New Workbook

Residuals

☐ Residuals ☐ Residual Plots

☐ Standardized Residuals ☐ Line Fit Plots

Normal Probability

☐ Normal Probability Plots

OK Cancel Help

Example – Compressor Map

Run the regression using all of the AHRI terms

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.999				
R Square	0.998				
Adjusted R Square	0.886				
Standard Error	239.8				
Observations	18				
ANOVA					
	df	SS	MS	F	Significance F
Regression	9	320106936	35567437	696.0	0.000
Residual	9	517417	57491		
Total	18	320624353			
	Coefficients	Std Error	t Stat	P-value	
Intercept	-154778.1	118080.4	-1.31	0.222	
Ts	0	0	65535.00	#NUM!	
Td	5426.5	3402.7	1.59	#NUM!	
Ts ²	141.1	58.9	2.40	0.040	
TsTd	-125.0	56.1	-2.23	0.053	
Td ²	-23.47	27.52	-0.85	0.416	
Ts ³	-1.25	0.67	-1.86	0.096	
Ts ² Td	0.522	0.453	1.15	0.279	
TsTd ²	0.337	0.157	2.14	0.061	
Td ³	0.0193	0.084	0.23	0.823	

10 AHRI terms

Example – Compressor Map

Run the regression using all of the AHRI terms

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.999				
R Square	0.998				
Adjusted R Square	0.886				
Standard Error	239.8				
Observations	18				
ANOVA					
	df	SS	MS	F	Significance F
Regression	9	320106936	35567437	696.0	0.000
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Intercept	-154778.1	118080.4	-1.31	0.222	
Ts	0	0	65535.00	#NUM!	
Td	5426.5	3402.7	1.59	#NUM!	
Ts ²	141.1	58.9	2.40	0.040	
TsTd	-125.0	56.1	-2.23	0.053	
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Td ³	0.0193	0.084	0.23	0.823	

10 AHRI terms

Failed to compute!

Example – Compressor Map

Why?

Consider how the input values lie across the input domain.

The terms need to be orthogonal and cover the space.



Example – Compressor Map

Why?

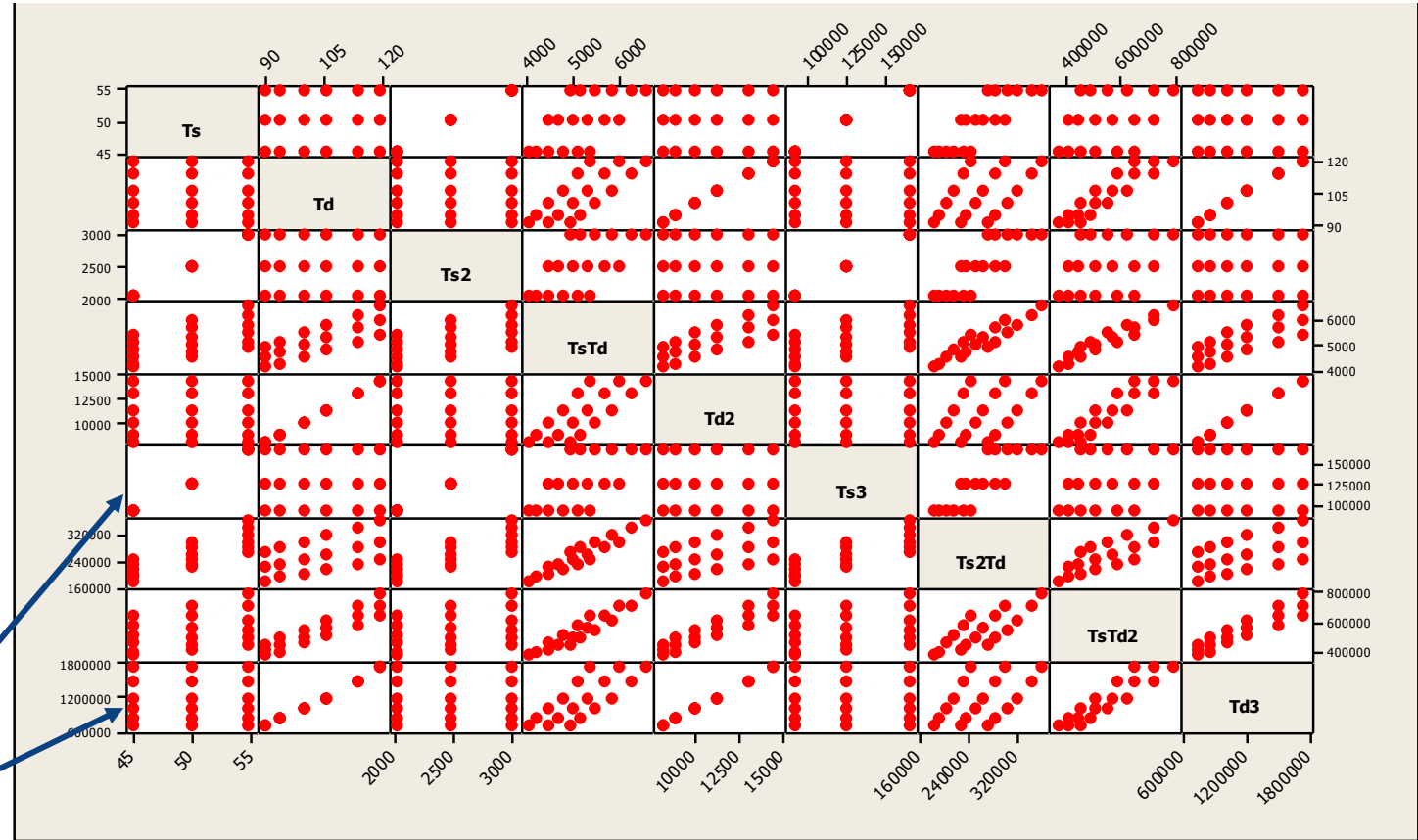
Consider how the input values lie across the input domain.

The terms need to be orthogonal and cover the space.

No collinear inputs!

A problem!

Good!



Example – Compressor Map

The problem was correlation among the input variables!

	T_s	T_d	T_s^2	$T_s T_d$	T_d^2	T_s^3	$T_s^2 T_d$	$T_s T_d^2$	T_d^3
T_s	1.000								
T_d	0.000	1.000							
T_s^2	1.000	0.000	1.000						
$T_s T_d$	0.625	0.778	0.624	1.000					
T_d^2	0.000	0.999	0.000	0.778	1.000				
T_s^3	0.998	0.000	1.000	0.624	0.000	1.000			
$T_s^2 T_d$	0.844	0.529	0.844	0.944	0.529	0.844	1.000		
$T_s T_d^2$	0.371	0.925	0.371	0.956	0.925	0.371	0.809	1.000	
T_d^3	0.000	0.997	0.000	0.776	0.999	0.000	0.528	0.925	1.000

Correlation between any two input terms

With this data, you can fit only one of T_s, T_s^2, T_s^3 .

With this data, you can fit only one of $T_d, T_d^2, T_s T_d^2, T_d^3$.

This happens *all the time* with field collected natural non-DOE data.

Example – Compressor Map

T_s and T_s^3 are highly correlated.

Re-run regression without T_s^3 :

9 AHRI terms

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.999				
R Square	0.998				
Adjusted R Square	0.997				
Standard Error	239.8				
Observations	18				
ANOVA					
	df	SS	MS	F	Significance F
Regression	8	3.2E+08	4.00E+07	696.0	0.000
Residual	9	5.17E+05	5.75E+04		
Total	17	3.21E+08			
	Coefficients	Std Error	t Stat	P-value	
Intercept	-309912	174166	-1.78	0.109	
Ts	9370.7	5039.8	1.86	0.096	
Td	5426.5	3402.7	1.59	0.145	
Ts ²	-46.95	47.37	-0.99	0.347	
TsTd	-125.01	56.09	-2.23	0.053	
Td ²	-23.47	27.52	-0.85	0.416	
Ts ² Td	0.522	0.453	1.15	0.279	
TsTd ²	0.337	0.157	2.14	0.061	
Td ³	0.019	0.084	0.23	0.823	

Example – Compressor Map

There are still many terms with high p-values.

Let's drop them from the regression.

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.999				
R Square	0.998				
Adjusted R Square	0.997				
Standard Error	239.8				
Observations	18				
ANOVA					
	df	SS	MS	F	Significance F
Regression	8	3.2E+08	4.00E+07	696.0	0.000
Residual	9	5.17E+05	5.75E+04		
Total	17	3.21E+08			
	Coefficients	Std Error	t Stat	P-value	
Intercept	-309912	174166	-1.78	0.109	
Ts	9370.7	5039.8	1.86	0.096	
Td	5426.5	3402.7	1.59	0.145	
Ts ²	-46.95	47.37	-0.99	0.347	
TsTd	-125.01	56.09	-2.23	0.053	
Td ²	-23.47	27.52	-0.85	0.416	
Ts ² Td	0.522	0.453	1.15	0.279	
TsTd ²	0.337	0.157	2.14	0.061	
Td ³	0.019	0.084	0.23	0.823	

Example – Compressor Map

Drop out the terms with high p-values

- r^2 went down by 0.001
- s grew to 278.9

But p-values changed and two terms are now insignificant

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.998				
R Square	0.997				
Adjusted R Square	0.996				
Standard Error	278.9				
Observations	18				
ANOVA					
	df	SS	MS	F	Significance F
Regression	4	3.20E+08	7.99E+07	1030	0.000
Residual	13	1.01E+06	7.78E+04		
Total	17	3.21E+08			
	Coefficients	Std Error	t Stat	P-value	
Intercept	18214.5	7979.4	2.28	0.040	
Ts	908.6	227.3	4.00	0.002	
Td	-154.6	76.3	-2.02	0.064	
TsTd	-0.273	3.477	-0.08	0.939	
TsTd ²	-0.0085	0.0149	-0.57	0.579	

Example – Compressor Map

Drop out the next terms with high p-values

- r^2 did not change
- s went down to 268.8

Again a term became insignificant...

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.998				
R Square	0.997				
Adjusted R Square	0.996				
Standard Error	268.8				
Observations	18				
ANOVA					
	df	SS	MS	F	Significance F
Regression	3	3.20E+08	1.07E+08	1470	0.000
Residual	14	1.01E+06	7.23E+04		
Total	17	3.21E+08			
	Coefficients	Std Error	t Stat	P-value	
Intercept	18486.5	6929.2	2.67	0.018	
Ts	891.7	70.4	12.67	0.000	
Td	-157.2	66.2	-2.37	0.032	
TsTd ²	-0.0095	0.0063	-1.52	0.152	

Example – Compressor Map

Drop out the next terms with high p-values

- r^2 did not change
- s went up to 280.2

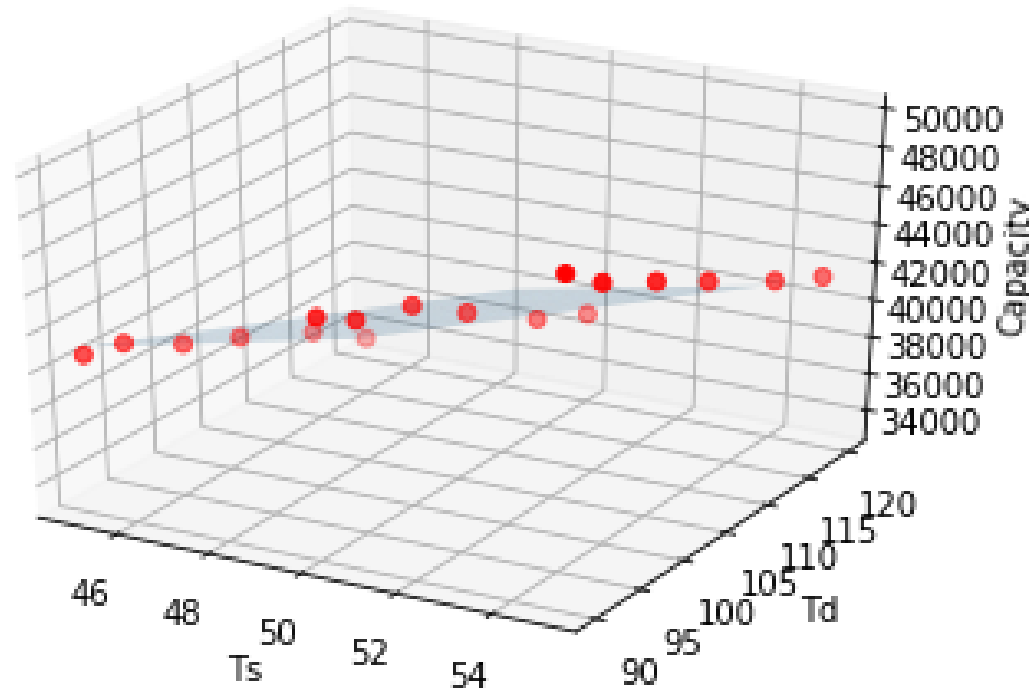
The linear model fit to r^2 over 99%

Adding other polynomial terms did nothing...

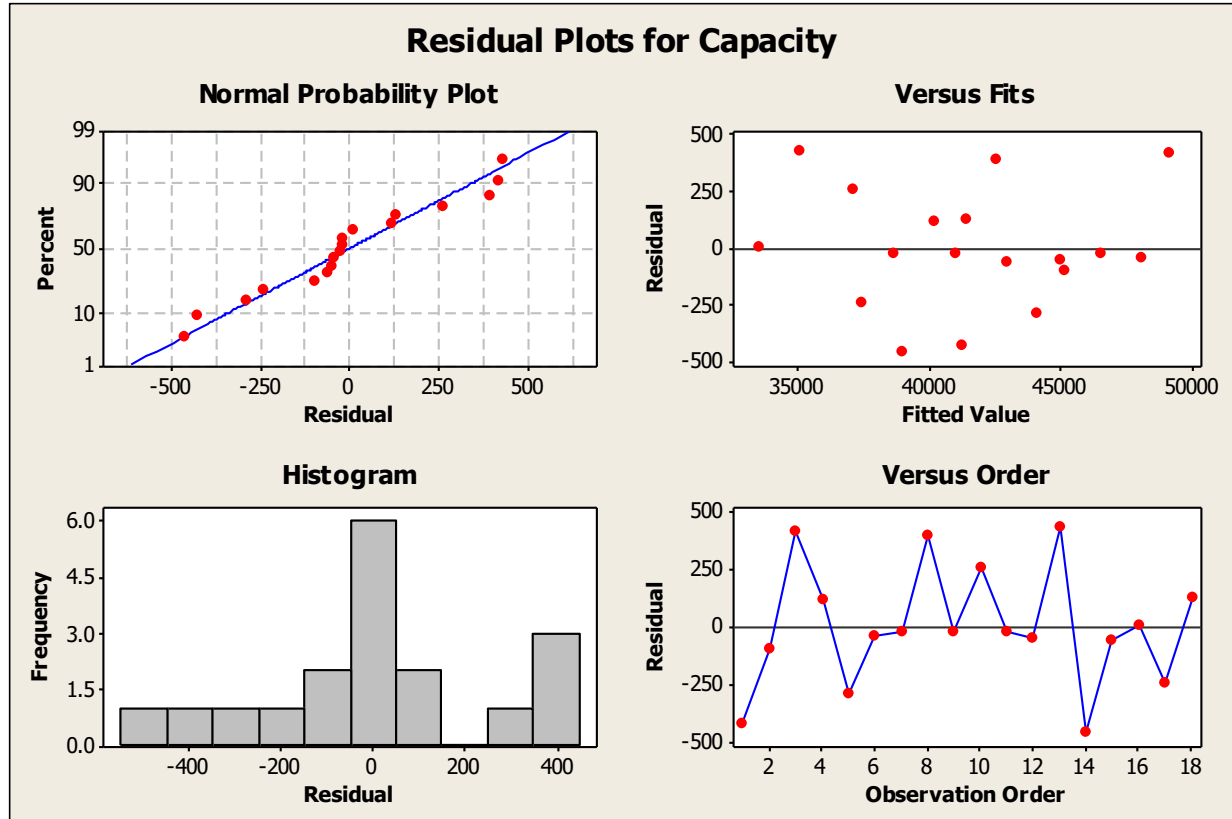
SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.998				
R Square	0.996				
Adjusted R Square	0.996				
Standard Error	280.2				
Observations	18				
ANOVA					
	df	SS	MS	F	Significance F
Regression	2	3.19E+08	1.60E+08	2030	0.000
Residual	15	1.18E+06	7.85E+04		
Total	17	3.21E+08			
	Coefficients	Std Error	t Stat	P-value	
Intercept	28887.5	1039.3	27.80	0.000	
Ts	787.6	16.2	48.68	0.000	
Td	-257.2	6.2	-41.21	0.000	

$$Y = 28,887.5 + 787.5T_s + 257.3T_d + 0T_s^2 + 0T_sT_d + 0T_d^2 + 0T_s^3 + 0T_s^2T_d + 0T_sT_d^2 + 0T_d^3$$

Example – Compressor Map



Example – Compressor Map





Comparing Models

You can often have two or three good competing models

Be biased toward simpler models. Look at the p-values to ensure added terms are significant

Models that make physical sense are preferred. Derive the physical relationship first and fit unknown constants.



Advanced: VIF

An indicator of how collinear a variable is with any other is indicated by the **Variance Inflation Factor** (VIF). A $VIF > 10$ indicates collinearity of that variable with another.

Coefficients

Term	Coef	SE Coef	95% CI	T-Value	P-Value	VIF
Constant	-42.8	46.0	(-161.1, 75.6)	-0.93	0.396	
Temp	0.394	0.313	(-0.410, 1.197)	1.26	0.263	19.00
Pressure	0.894	0.551	(-0.522, 2.309)	1.62	0.166	59.00
Conc	0.18	1.47	(-3.59, 3.96)	0.12	0.906	59.00
Temp*Pressure	-0.00219	0.00358	(-0.01140, 0.00703)	-0.61	0.568	59.00
Temp*Conc	0.00083	0.00956	(-0.02374, 0.02541)	0.09	0.934	59.00
Pressure*Conc	-0.00250	0.00956	(-0.02707, 0.02207)	-0.26	0.804	19.00

Two or more of these inputs are correlated.

Minitab – Fit the Model

Regression

Responses:
Capacity

Continuous predictors:
Ts Td

Categorical predictors:

Model... Options... Coding... Stepwise...

Validation... Graphs... Results... Storage...

Select

Help

OK Cancel

Regression: Model

Predictors:
Ts
Td

Add terms using selected predictors and model terms:

Interactions through order: 2 Add

Terms through order: 3 Add

Cross predictors and terms in the model Add

Terms in the model:

Default X Down Up

Ts
Td
Ts*Ts
Td*Td
Ts*Td
Ts*Ts*Ts
Td*Td*Td
Ts*Ts*Td
Ts*Td*Td

☒ Include the constant term in the model

Help

OK Cancel

WORKSHEET 1

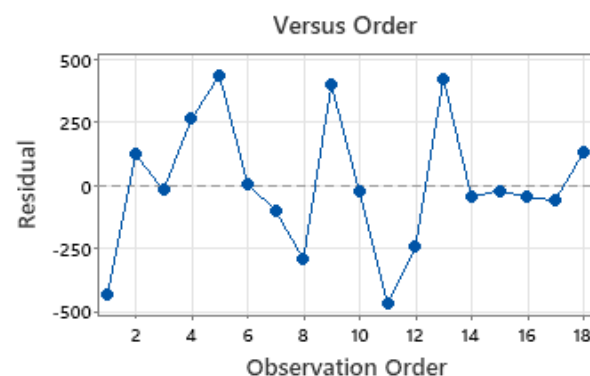
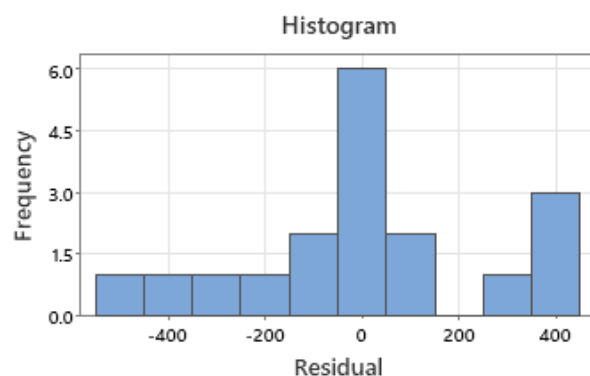
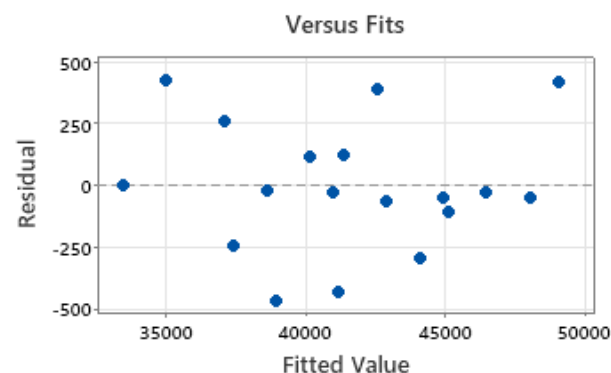
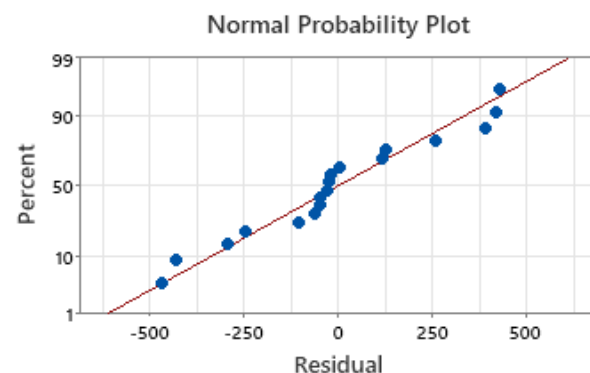
Regression Analysis: Capacity ver

The following terms cannot be estima
 $Ts*Ts*Ts$

Regression Equation

Capacity = $-310056 + 9373 Ts +$
 $+ 0.0194 Td*Td*Td + 0.522 Ts*Ts*Td$

Residual Plots for Capacity





Python – Fit the Model



OLS Summary

Regression is a means to find a best set of equation variables to fit a set of data

Ordinary least squares minimizes the errors in response, with perfect inputs

Regression can be used with natural data only when inputs are independent

For more general problems with errors in both X and Y then maximum likelihood estimation is required



Overview

OLS regression

Regression Table

Residual Analysis

Polynomial Regression

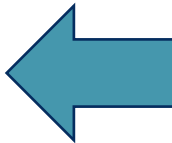
Multiple Regression

Over Fitting

Maximum Likelihood Regression

Fitting a Distribution Function

Suppose we have a data set of n observed values. We believe they should follow a particular distribution function, perhaps a Weibull distribution function:

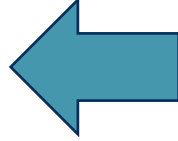
$$f(t) = \frac{\beta}{\alpha} \left(\frac{t}{\alpha}\right)^{\beta-1} e^{-\left(\frac{t}{\alpha}\right)^{\beta}}$$


Given the data points $(t_1, t_2, \dots, t_n)$, what are values of (α, β) that best fit the data?

Life (yrs)
18.83
20.80
21.68
23.03
23.23
24.05
24.32
25.50
25.52
25.80
26.69
26.77
26.78
27.05
27.67
29.90
31.11
33.20
33.73
33.76
33.89
34.76
35.75
35.91
36.98
37.08
37.09
39.58
44.05
45.29
45.38

Fitting a Distribution Function

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$$f(t) = \frac{\beta}{\alpha} \left(\frac{t}{\alpha}\right)^{\beta-1} e^{-\left(\frac{t}{\alpha}\right)^{\beta}}$$


Given the data points $(t_1, t_2, \dots, t_n)$, what are values of (α, β) that best fit the data?

More generally, suppose the distribution density function is given by $f(t, \theta)$ where θ are the parameters of the distribution.

Life (yrs)
18.83
20.80
21.68
23.03
23.23
24.05
24.32
25.50
25.52
25.80
26.69
26.77
26.78
27.05
27.67
29.90
31.11
33.20
33.73
33.76
33.89
34.76
35.75
35.91
36.98
37.08
37.09
39.58
44.05
45.29
45.38



Maximum Likelihood Estimation

Consider a distribution density function is given by $f(x, \theta)$ where θ are the parameters of the distribution.

The data points $(x_1, x_2, \dots, x_n)$ were observed from random variables $(X_1, X_2, \dots, X_n)$, where each X_i is given by a distribution function $f(\theta)$.

We would like to find θ which maximizes this probability of our data, since that would give us belief that this distribution function could have produced this data.



Maximum Likelihood Estimation

Consider a distribution density function is given by $f(x, \theta)$ where θ are the parameters of the distribution.

The data points $(x_1, x_2, \dots, x_n)$ were observed from random variables $(X_1, X_2, \dots, X_n)$, where each X_i is given by a distribution function $f(\theta)$.

We would like to find θ which maximizes this probability of our data, since that would give us belief that this distribution function could have produced this data.

The probability of attaining the values observed is, for a small ε around each point,

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = \varepsilon^n f(x_1, \theta) f(x_2, \theta) \cdots f(x_n, \theta)$$

Lets find θ which maximizes this P .



Maximum Likelihood Estimation

Given a proposed probability density $f(x, \theta)$, then define the *likelihood function* as

$$L(\theta) = f(x_1, \theta)f(x_2, \theta) \cdots f(x_n, \theta)$$

This gives how probable the data is given θ .

The estimate θ^* which maximizes $L(\theta)$ we will call the *maximum likelihood estimate*.

$$\text{Find } \theta^* = \underset{\theta}{\operatorname{argmax}} L(\theta)$$

Finding these best parameter estimates for the distribution function is in general an optimization problem.



Maximum Likelihood Estimation

The *likelihood function* and the *probability density function* are the same thing really.

With known parameters θ and unknown event inputs x , we call it a probability function.

$$f(x, \theta)$$

With unknown parameters θ and known sample inputs x_i , we call it a likelihood function.

$$L(\theta) = f(x_1, \theta)f(x_2, \theta) \cdots f(x_n, \theta)$$



Maximum Likelihood Estimation

The likelihood function is a multiplication of the $f(x_i)$ values, which can be numerically difficult to use when solving for θ .

$$L(\theta) = f(x_1, \theta)f(x_2, \theta) \cdots f(x_n, \theta)$$

Taking the $\ln()$ converts this into a summation.

$$LL(\theta) = \ln f(x_1, \theta) + \cdots + \ln f(x_n, \theta)$$

The *log-likelihood* function is a more well behaved summation and so typically used.

For some problems this can be solved using calculus. Otherwise, statistical code libraries apply numerical methods to solve for the maximum likelihood estimates.



Example: Binomial Distribution

Suppose a factory is producing thousands of parts per day. Yesterday 200 were sampled, and 12 were defective. Given that defectives follows a binomial distribution, what is best estimate of the defective rate π given the observed sample data defective rate p ?

Intuitively, the best estimate for π is the sample defective rate $p = \frac{c}{n} = \frac{12}{200} = 0.06$ or 6%.

Using MLE: the count c we will observe in n trial follows the binomial distribution with parameters $\theta = (n, p)$, where

$$f(c, n, p) = \binom{n}{c} p^c (1 - p)^{n-c}$$

What value of p best fits this function when $c = 12$ and $n = 200$?

Example: Binomial Distribution

Constructing the likelihood function,

$$L(p) = \binom{200}{12} p^{12} (1 - p)^{200-12}$$

What value of p maximizes this function when $c = 12$ and $n = 200$?

Taking the $\log(L(p))$,

$$\log(L(p)) = \log\left(\binom{200}{12}\right) + 12\log p + 188\log(1 - p)$$

Then differentiate:

$$\frac{\partial}{\partial p} \log(L(p)) = 0 = 12 \frac{1}{p} - 188 \frac{1}{1 - p}$$

Solving, $p = \frac{12}{200} = 6\%$. The sample defective rate is the best estimate.

Example: Weibull Distribution

Many failure modes' time-to-failure are well represented using a 2 parameter Weibull distribution given by $f(t) = \frac{\beta}{\alpha} \left(\frac{t}{\alpha}\right)^{\beta-1} e^{-\left(\frac{t}{\alpha}\right)^\beta}$. The parameters are $\theta = (\alpha, \beta)$.

The likelihood function is give by

$$L(\alpha, \beta) = \prod_{i=1}^n f(\alpha, \beta, t_i) = \prod_{i=1}^n \frac{\beta}{\alpha} \left(\frac{t_i}{\alpha}\right)^{\beta-1} e^{-\left(\frac{t_i}{\alpha}\right)^\beta}$$

And the log-likelihood given by

$$L(\alpha, \beta) = n(\ln(\beta) - \beta \ln(\alpha)) + (\beta - 1) \sum_{i=1}^n \ln(t_i) - \sum_{i=1}^n \left(\frac{t_i}{\alpha}\right)^\beta$$

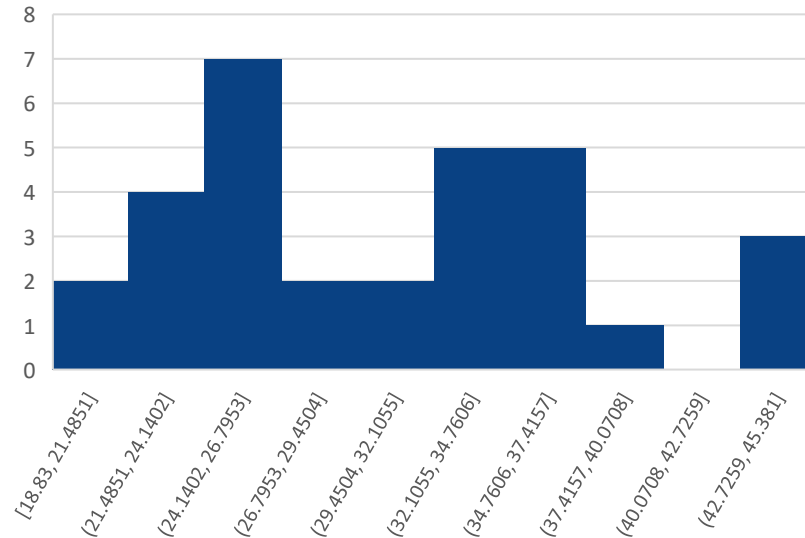
Example: Weibull Distribution

Consider the Fuller dataset from NIST showing the time to failure of airplane window glass, <https://www.itl.nist.gov/div898/handbook/datasets/FULLER2.DAT>

Assuming the window life is Weibull

distributed $f(t) = \frac{\beta}{\alpha} \left(\frac{t}{\alpha}\right)^{\beta-1} e^{-\left(\frac{t}{\alpha}\right)^\beta}$,

what are the best distribution statistics (α, β) to fit the data?



Life (yrs)
18.83
20.8
21.657
23.03
23.23
24.05
24.321
25.5
25.52
25.8
26.69
26.77
26.78
27.05
27.67
29.9
31.11
33.2
33.73
33.76
33.89
34.76
35.75
35.91
36.98
37.08
37.09
39.58
44.045
45.29
45.381

Example: Weibull Distribution

Solving this manually using Excel, use the solver to maximize the Log(L) cell by search over the alpha and beta cells.

Life data
 t_i

B	C	D
Life (yrs)	Exp(t/a)^b	Ln(t)
18.83	0.067579	2.935451
20.8	0.107181	3.034953
21.657	0.129241	3.075329
23.03	0.171848	3.136798
23.23	0.178876	3.145445
24.05	0.210082	3.180135
24.321	0.221282	3.19134
25.5	0.275577	3.238678
25.52	0.27658	3.239462
25.8	0.29093	3.250374
26.69	0.340457	3.284289
26.77	0.345213	3.287282
26.78	0.345811	3.287655
27.05	0.362272	3.297687
27.67	0.402397	3.320349
29.9	0.576357	3.397858
31.11	0	
33.2	0	
33.73		
33.76	1.011862	3.519277
33.89	1.03005	3.52312
34.76	1.158472	3.548467
35.75	1.319532	3.578467
35.91	1.34713	3.581467
36.98	1.543539	3.610977
37.08	1.562982	3.613078
37.09	1.564937	3.613347
39.58	2.114962	3.678324
44.045	3.47121	3.785212
45.29	3.949985	3.813086
45.381	3.986909	3.815094

$$= \exp(t_i/a)^b$$

$$= \ln(t_i)$$

n	31
beta	4.635382794
alpha	33.67422708
Likelihood	1.53724E-46
Log(L)	-105.488925

Find parameter estimates...

...that maximize this

$$= n * (\ln(\text{beta}) - \text{beta} * \ln(\text{alpha})) + (\text{beta} - 1) * \text{SUM}(\text{D17:D47}) - \text{SUM}(\text{C17:C47})$$

...we will return MLE estimation when we discuss reliability.



Summary

Ordinary least squares regression can be used to determine the best fit coefficients in a linear summation formula.

For fitting non-linear distribution functions, maximum likelihood estimation can be applied.

Data that comes from natural processes often has correlated inputs and a full set of model coefficients cannot be precisely fit with OLS. Advanced topics may be needed.

Next topic: Design of Experiments